Lecture 5: Introduction to RAG

Pavlos Protopapas



Logistics / Reminders

1. Approx. **25%** of students have groups - **Milestone 1 due 09/29**

   (Highly recommend forming your own groups, post ideas on Ed)

1. Office Hours see posts on Ed -

   (a) https://edstem.org/us/courses/97373/discussion/8242347

   (b) https://edstem.org/us/courses/97373/discussion/8258201

1. AWS Credits - https://edstem.org/us/courses/97373/discussion/8279148

1. When you form a team, please fill this form too.
   https://bit.ly/ac215_2026_groups

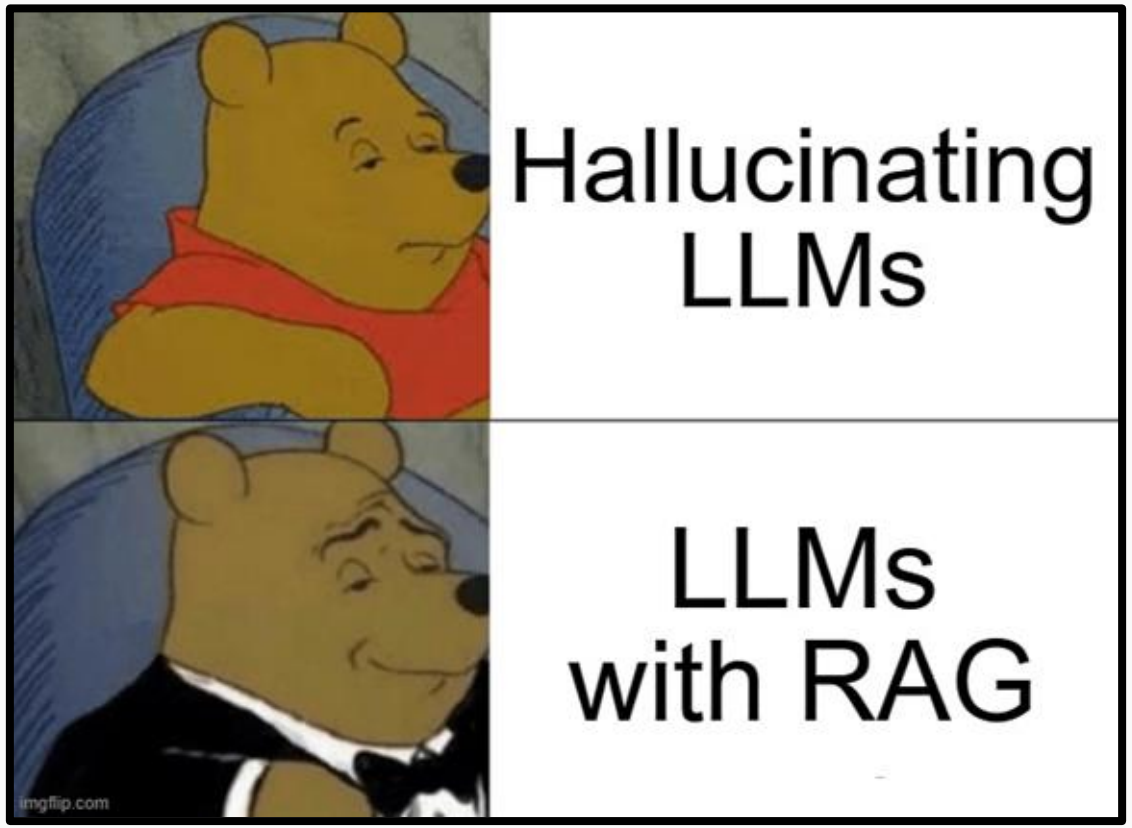
Outline

- Introduction - RAG
- Naïve RAG

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- **Introduction - RAG**
- Naïve RAG

Introduction - RAG



Introduction - RAG

LLMs ISSUES?

- Training data is not frequently updated
- Lack domain knowledge in specialized use cases
- Hallucinations
- Don’t cite the sources used to generate a response

Introduction - RAG

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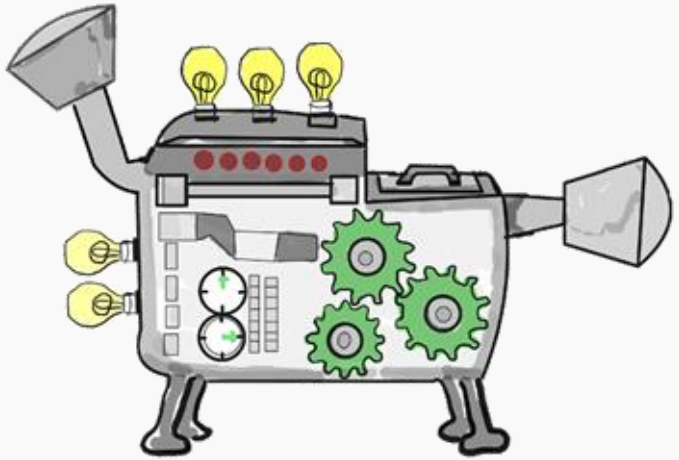
Let's try to overcome the first two issues with a new concept:
Retrieval-Augmented-Generation (RAG)

Introduction - RAG

If we have many documents, how can we process/query them using an LLM?

1. Pass all the text directly into an LLM





LLM

If the context is too big, the LLM gives out garbage.

Garbage

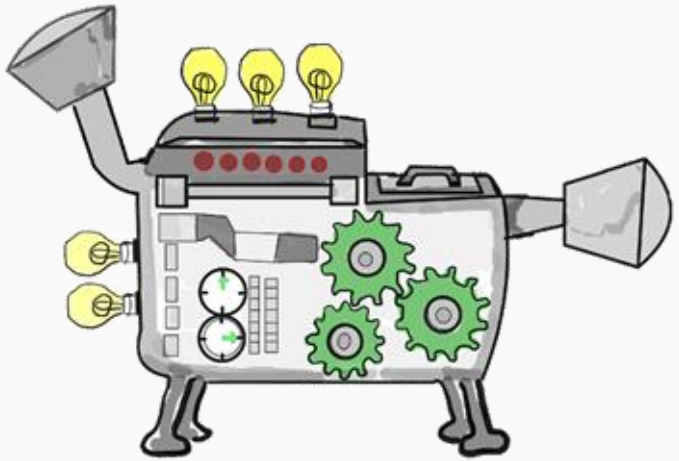
Introduction - RAG

If we have many documents, how can we process/query them using an LLM?

2. We can finetune our LLM using the data.

We will talk about fine-tuning in upcoming lectures.





LLM

Takes too long!

Introduction - RAG

If we have many documents, how can we process/query them using an LLM?

3. We can use RAG.

Let's take a deeper look at what RAG is and how it can help us.



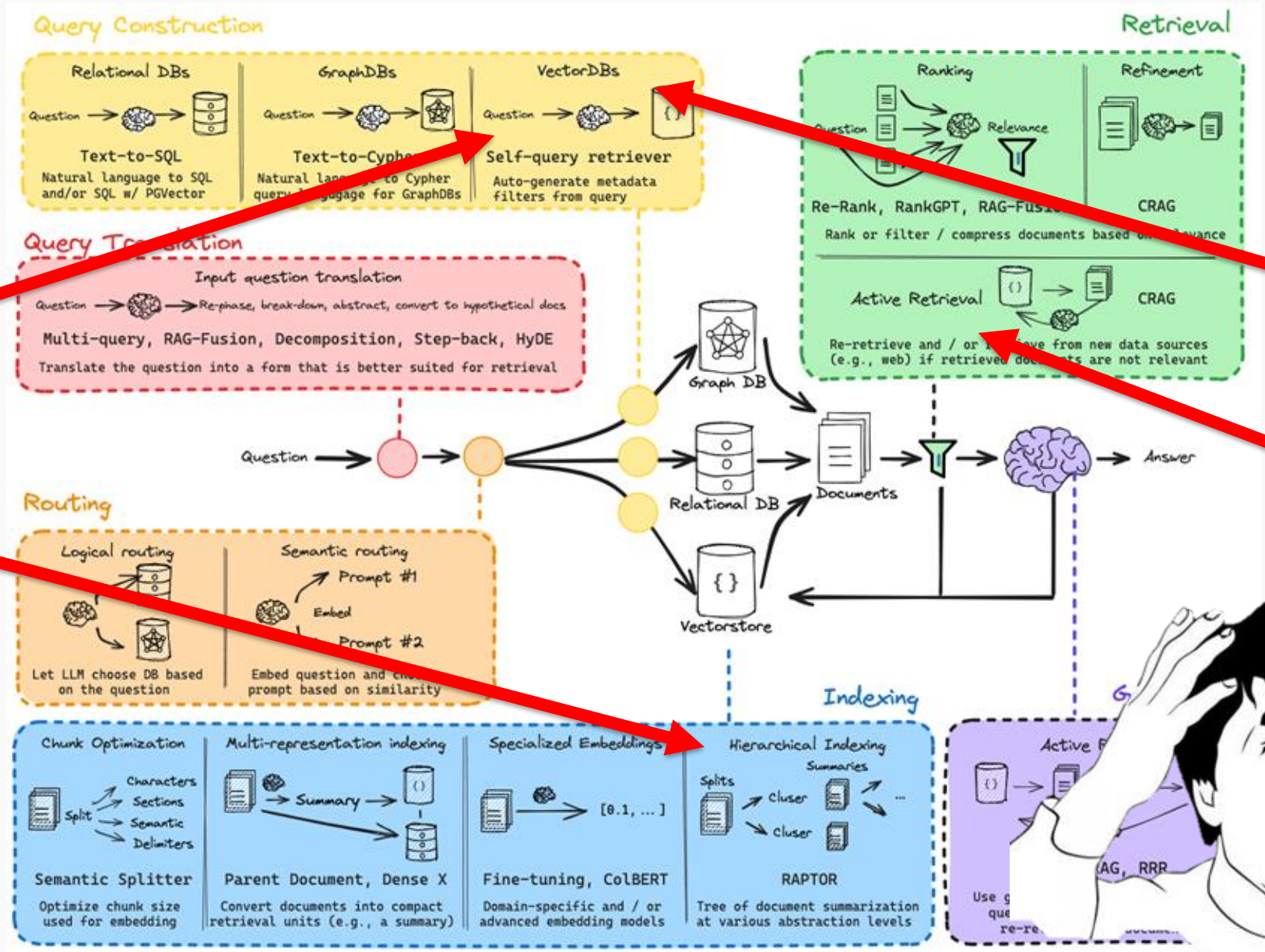
Retrieval Augmented Generation

What is Vector Database?

What is embedding?

What is query

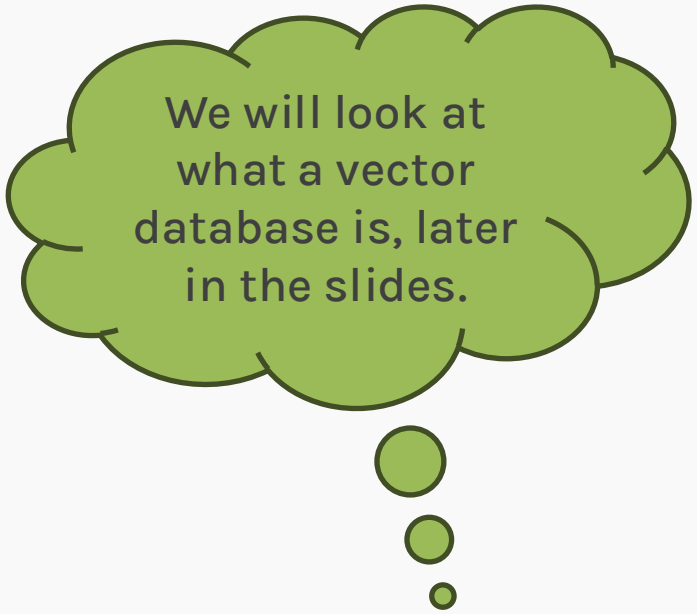
What is context?

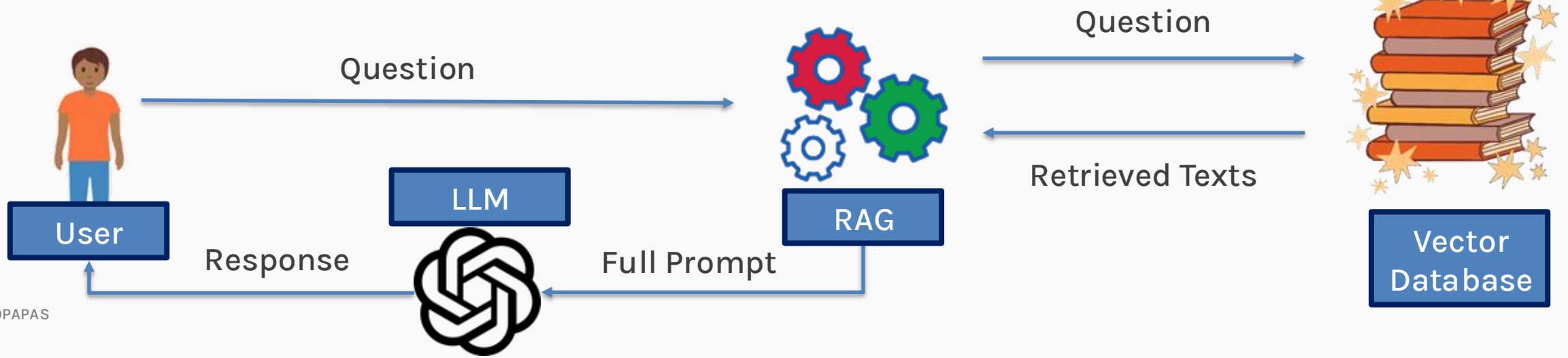


Introduction - RAG

What is RAG?

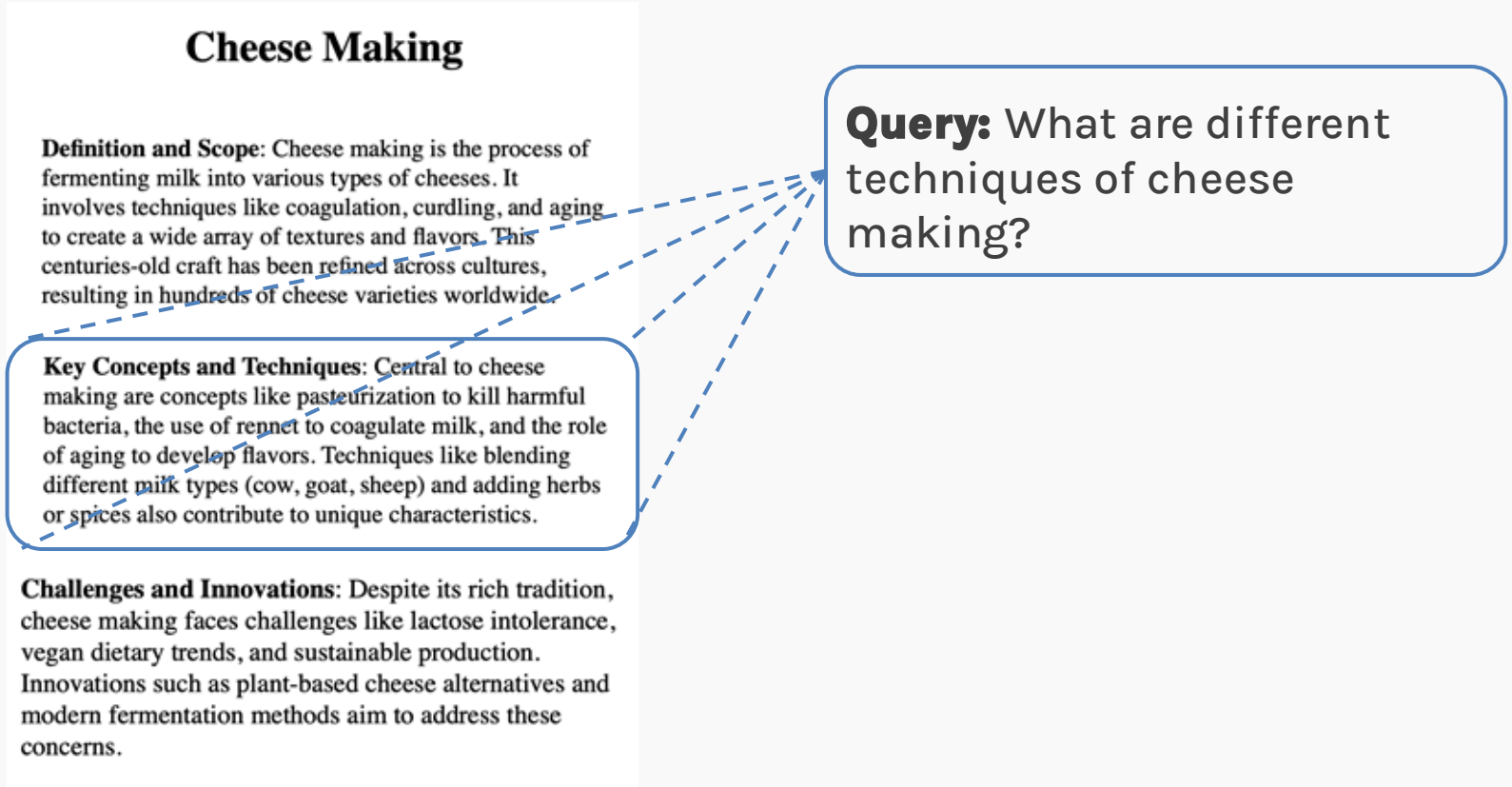
- RAG stands for Retrieval-Augmented-Generation.
- It is technique that improves the performance of a LLM, especially for tasks that require accurate and detailed information.
- A simple breakdown of how it works:





Introduction - RAG

What is RAG?



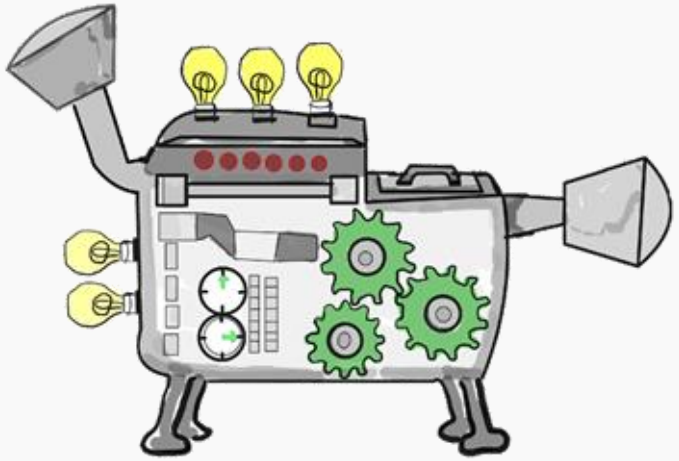
Introduction - RAG

What is RAG?

Query: What are different techniques of cheese making?

Key Concepts and Techniques: Central to cheese making are concepts like pasteurization to kill harmful bacteria, the use of rennet to coagulate milk, and the role of aging to develop flavors. Techniques like blending different milk types (cow, goat, sheep) and adding herbs or spices also contribute to unique characteristics.

Context



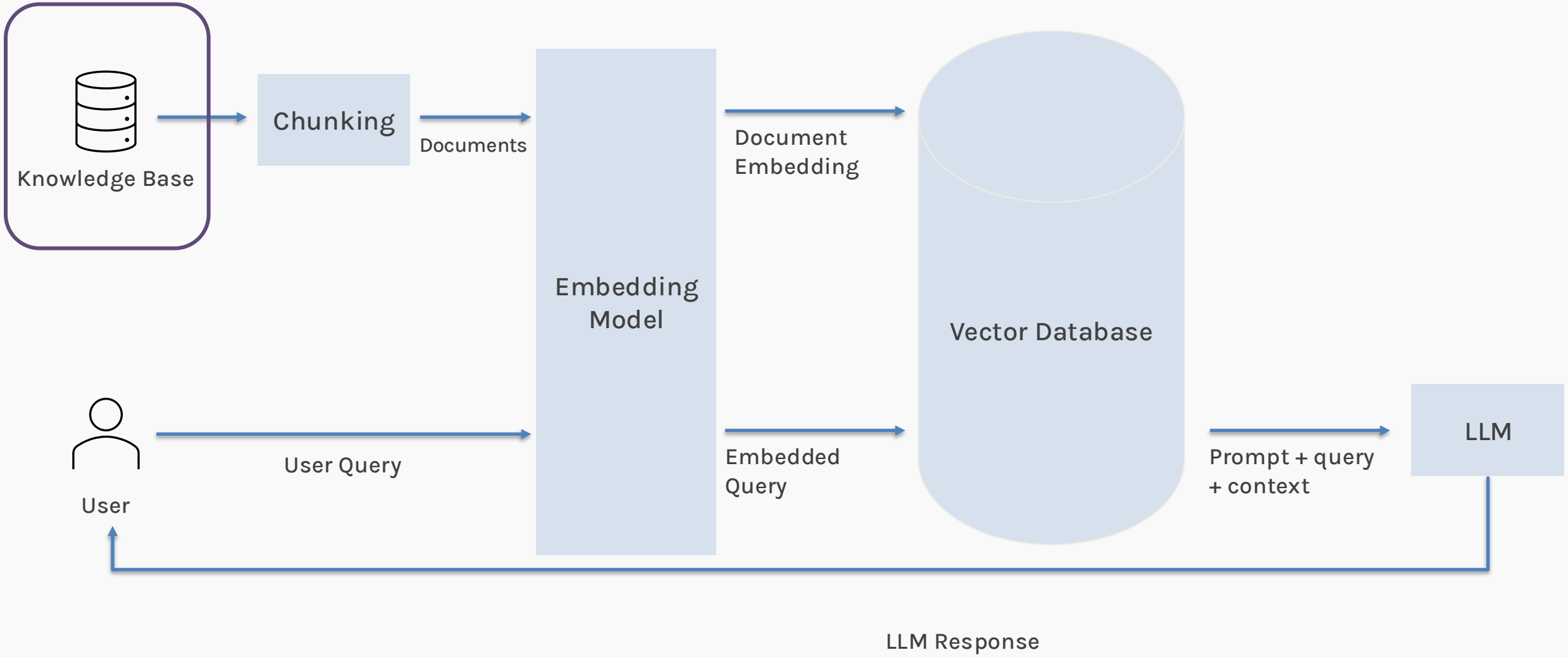
LLM

The different techniques of cheese making are ....

Outline

- Introduction - RAG
- **Naïve RAG**

Naïve RAG


Knowledge Base
Chunking
Documents
Embedding Model
Document Embedding
Vector Database
User
User Query
Embedded Query
Prompt + query + context
LLM
LLM Response

Naïve RAG – Knowledge Base

This could be any data source that contains useful information.



Websites



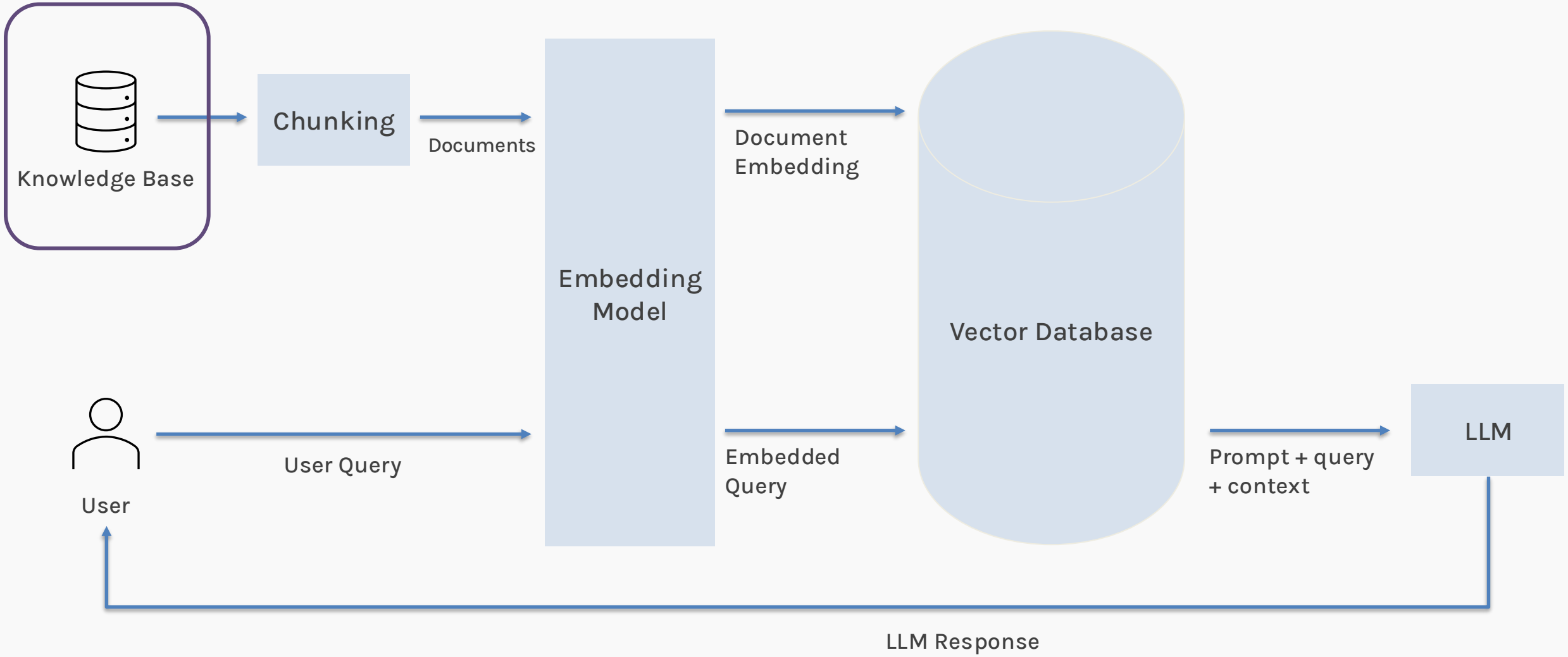
PDF Files



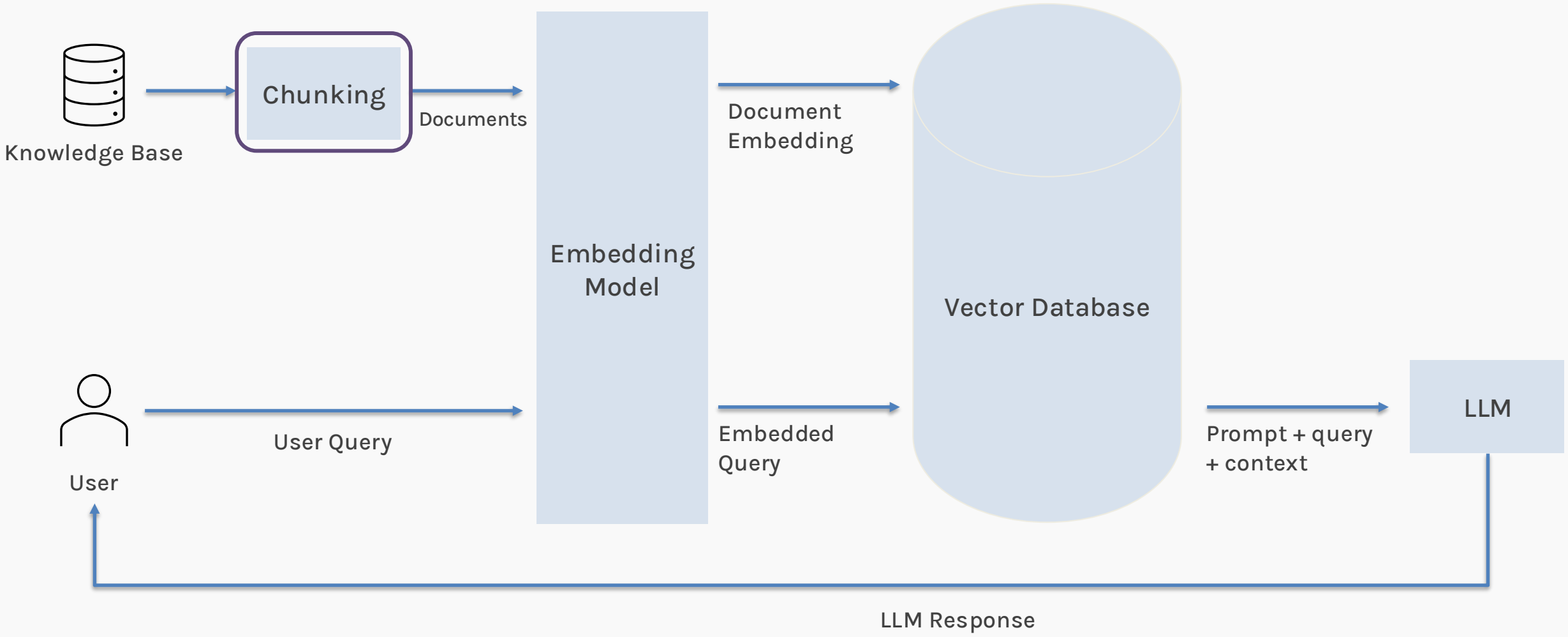
Word Files



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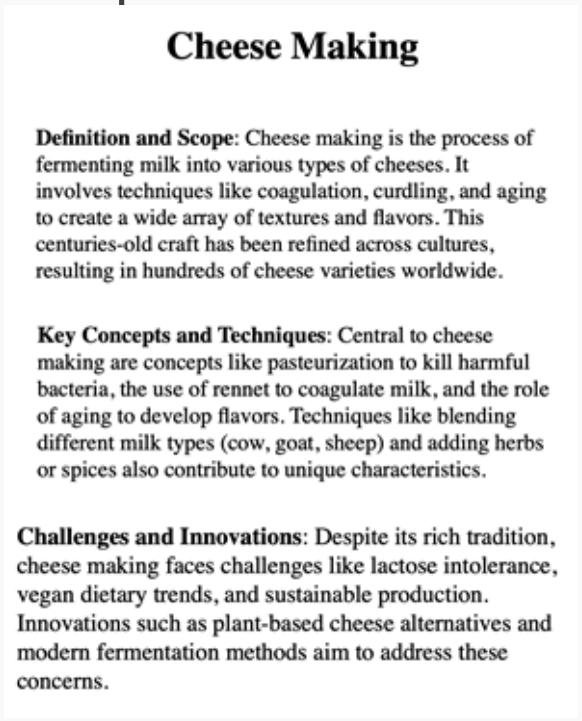
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Naïve RAG - Chunking

Chunking refers to the process of breaking down or segmenting information into smaller, more manageable pieces.

Chunking



Cheese Making

Definition and Scope: Cheese making is the process of fermenting milk into various types of cheeses. It involves techniques like coagulation, curdling, and aging to create a wide array of textures and flavors. This centuries-old craft has been refined across cultures, resulting in hundreds of cheese varieties worldwide.

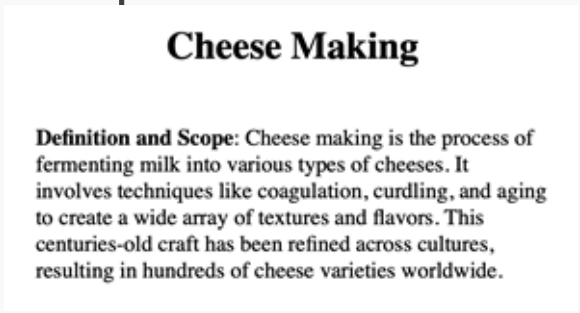
Key Concepts and Techniques: Central to cheese making are concepts like pasteurization to kill harmful bacteria, the use of rennet to coagulate milk, and the role of aging to develop flavors. Techniques like blending different milk types (cow, goat, sheep) and adding herbs or spices also contribute to unique characteristics.

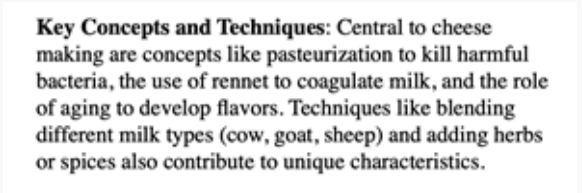
Challenges and Innovations: Despite its rich tradition, cheese making faces challenges like lactose intolerance, vegan dietary trends, and sustainable production. Innovations such as plant-based cheese alternatives and modern fermentation methods aim to address these concerns.

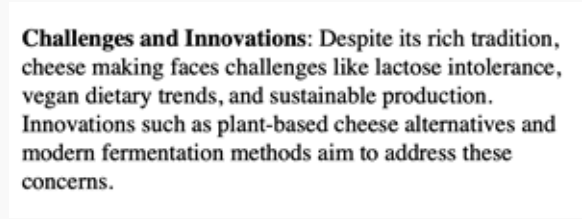
Naïve RAG - Chunking

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Chunking







This helps the model focus on the most relevant information when generating a response.

Naïve RAG - Chunking

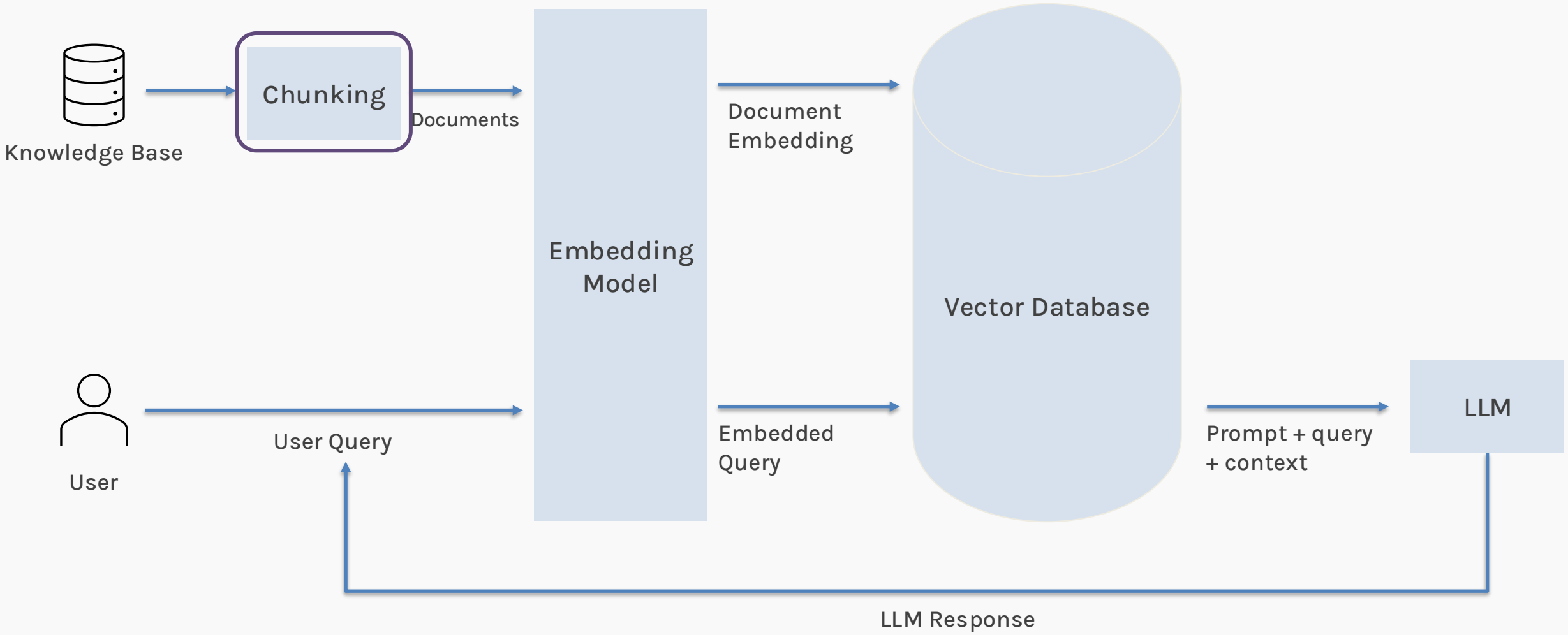
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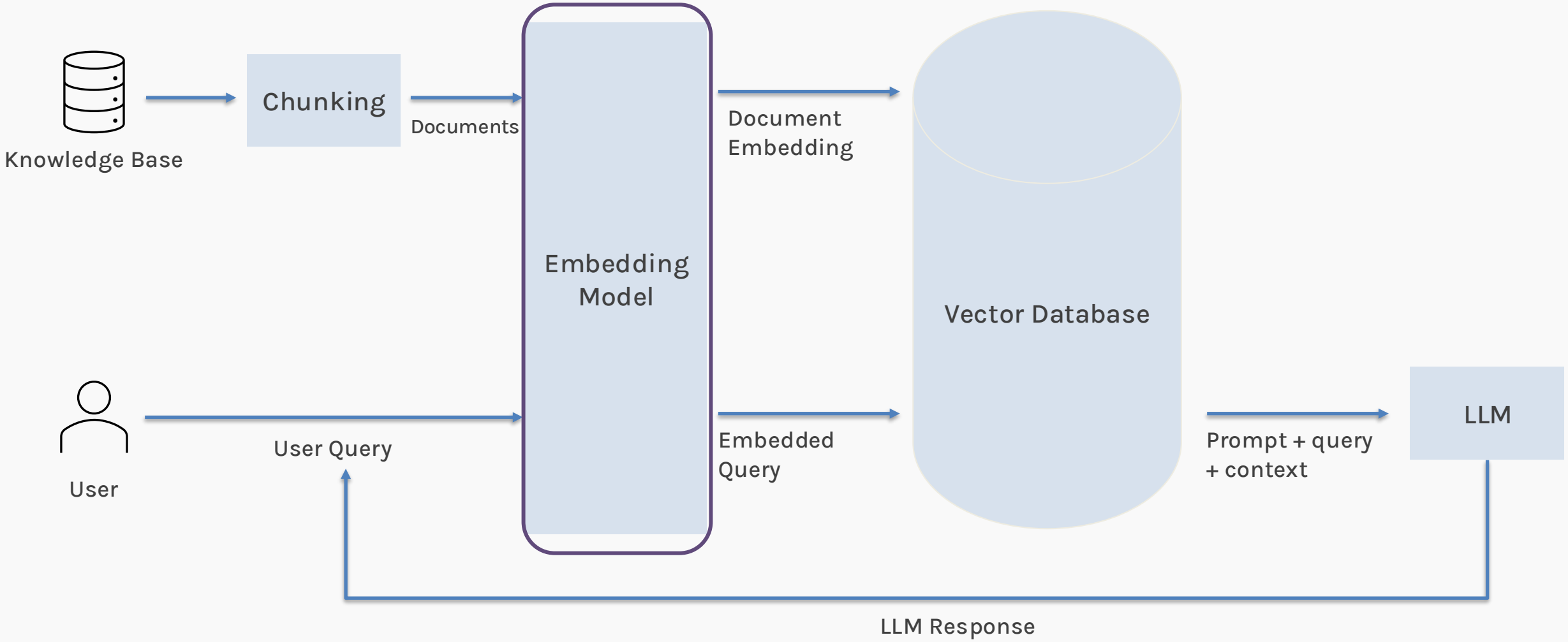
There are different methods of chunking which will be discussed later

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Naïve RAG

The embedding model converts text (docu
into vectors/embeddings that capture sem

We use an architecture similar to BERT and aggregate the word embeddings

Embedding Model

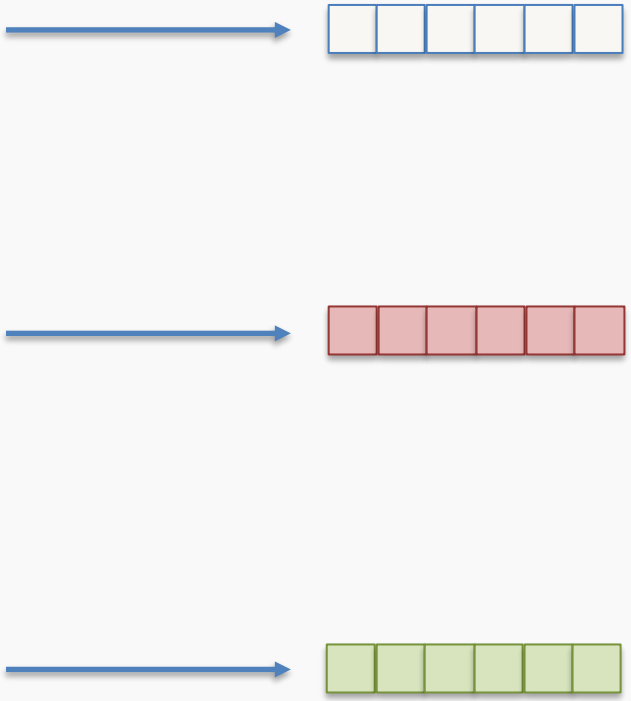
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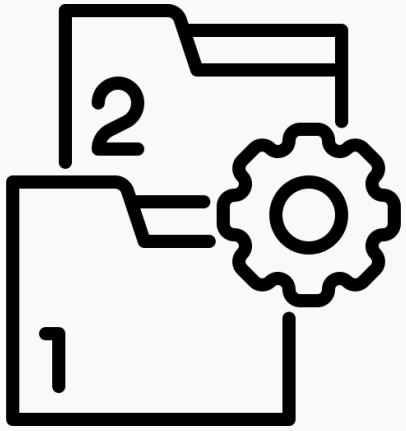
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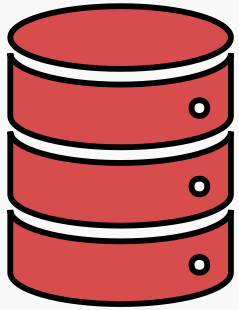
Embeddings



Index data

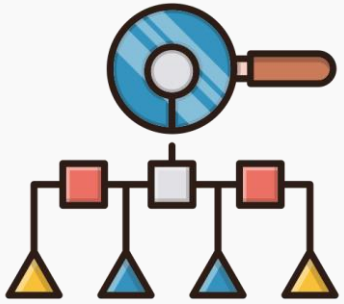
Index
Data



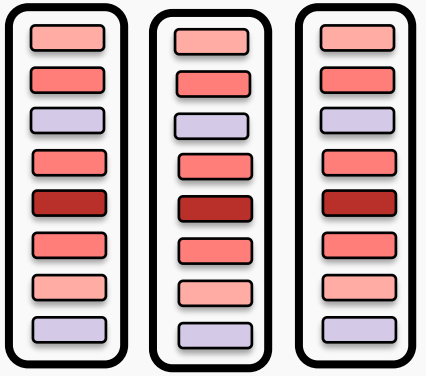




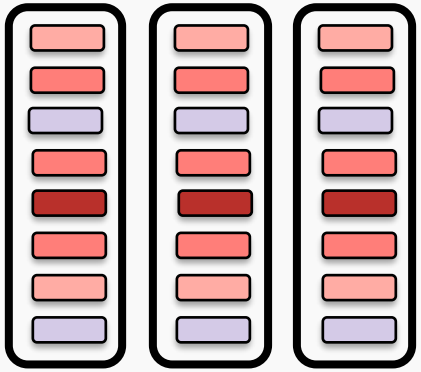
Lexical



Graphical



Semantic

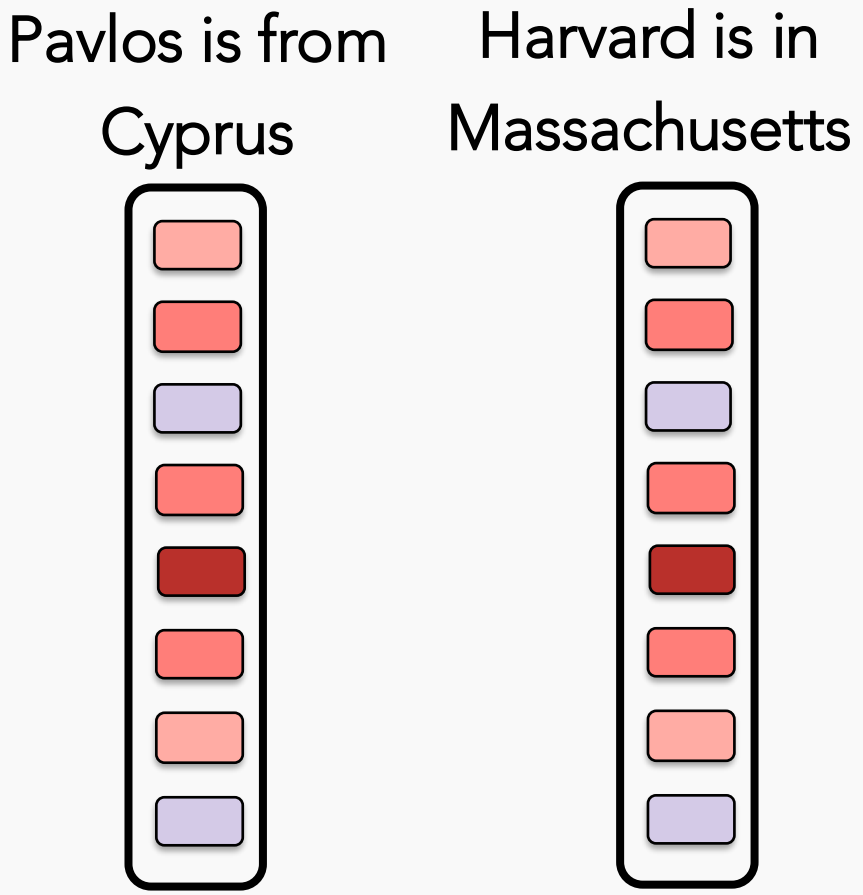


Semantic
Embeddings

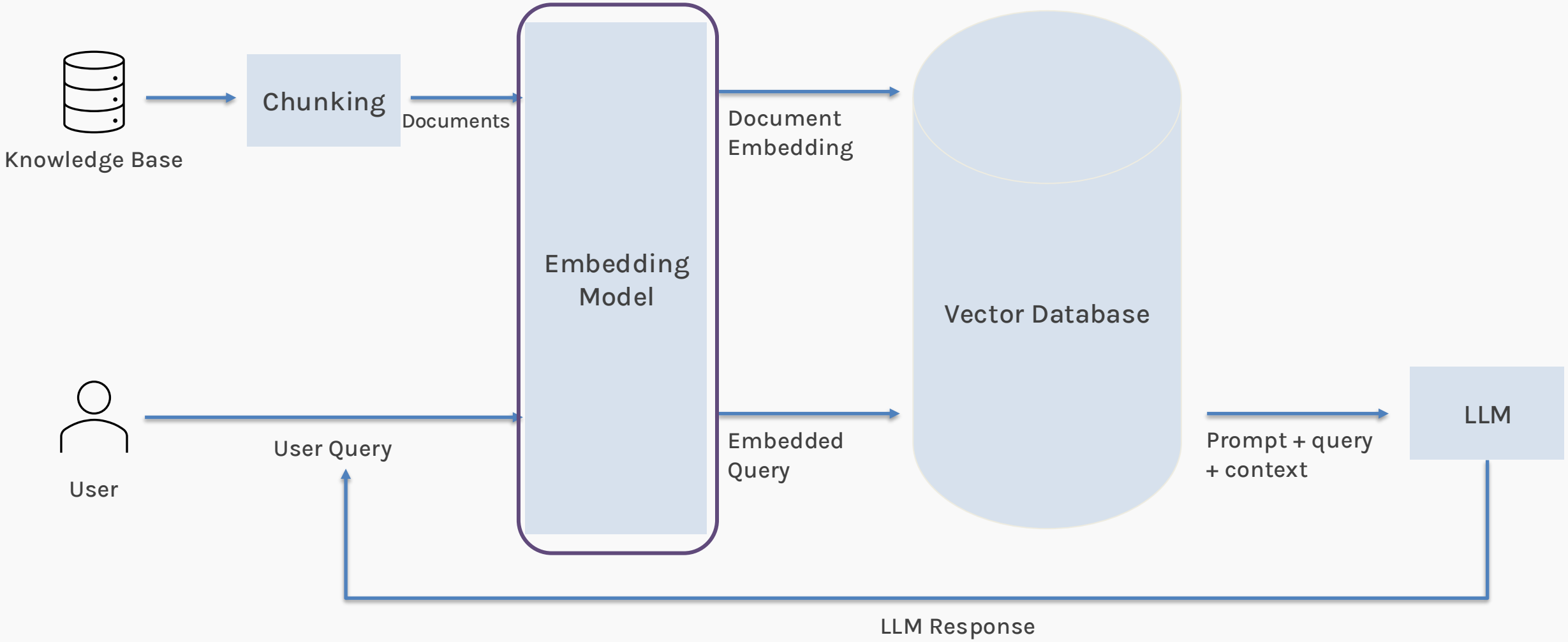
~~Word~~ Sentence Embeddings

- We use the same idea to map **sentences** in a vocabulary to a n-dimensional vector space.
- Such a vector is called a **Sentence Embedding**.
- Two sentences will be "**similar**" if their vector representations are ***close*** to each other.

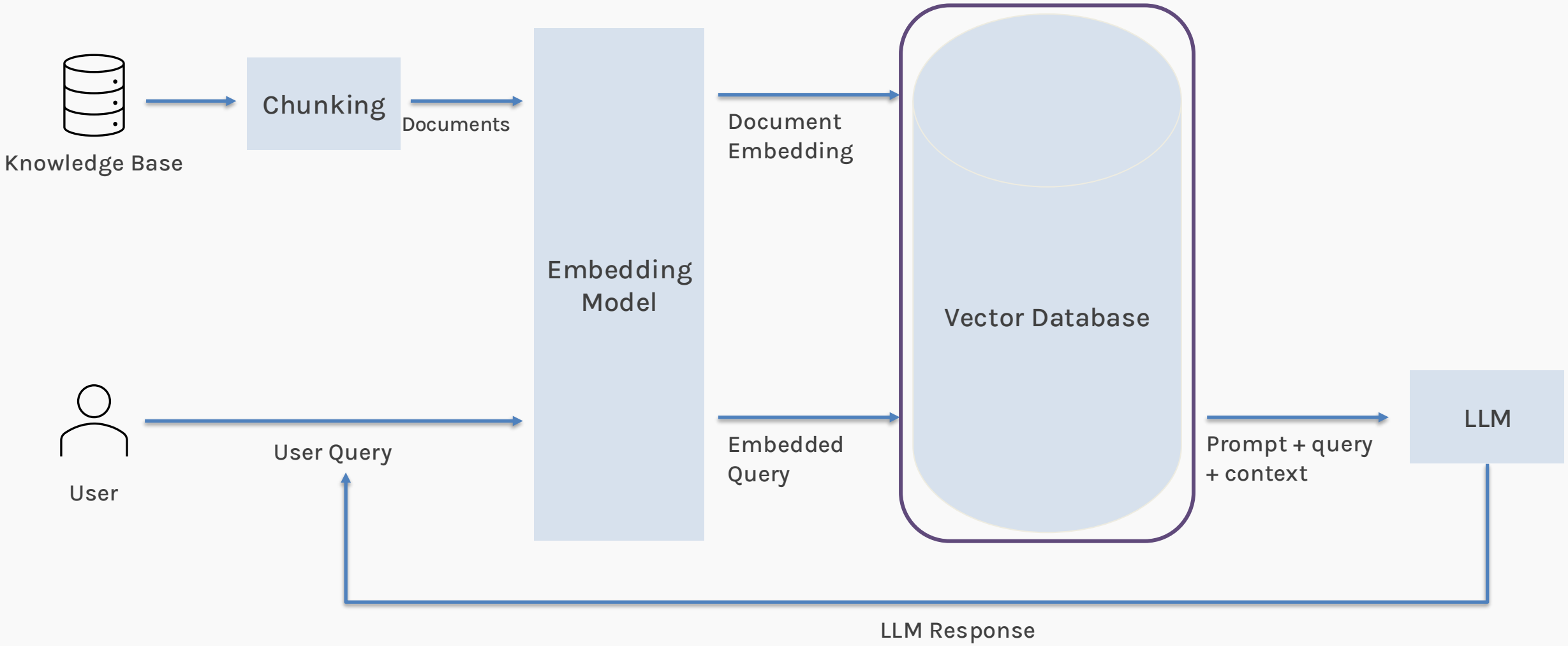
NOTE: A sentence embedding is different from individual word embeddings



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Naïve RAG – Vector Database



When it comes to RAG, we need to find information that's **semantically similar**

Traditional relational databases excel at:

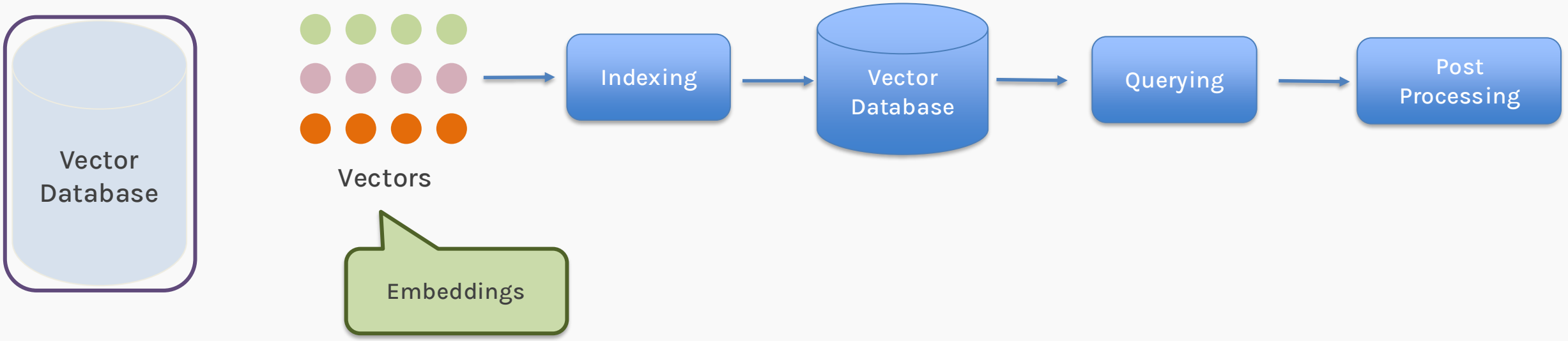
- Exact matches: WHERE user_id = 123
- Range queries: WHERE price BETWEEN 10 AND 50
- Structured data with clear relationships

But they struggle with semantic search - understanding that "best cheese for melting" and "cheeses Pavlos would choose for his midnight snack" mean similar things.

Vector databases solve this problem.

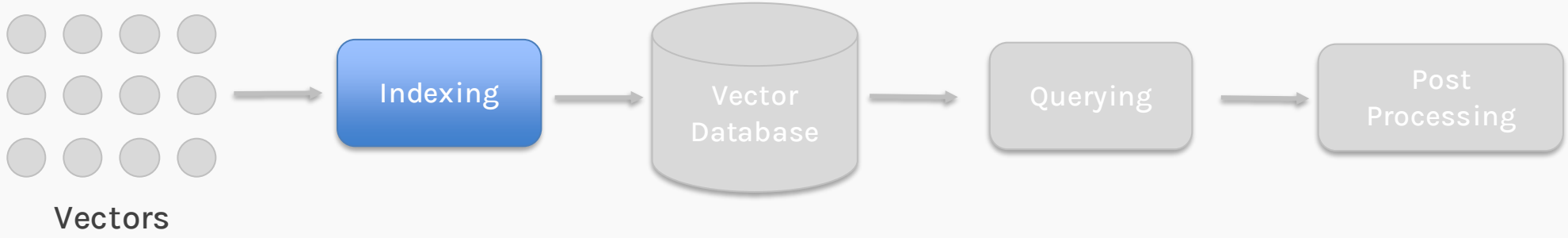
Naïve RAG – Vector Database

A vector database indexes and stores vector embeddings for fast retrieval and similarity search.



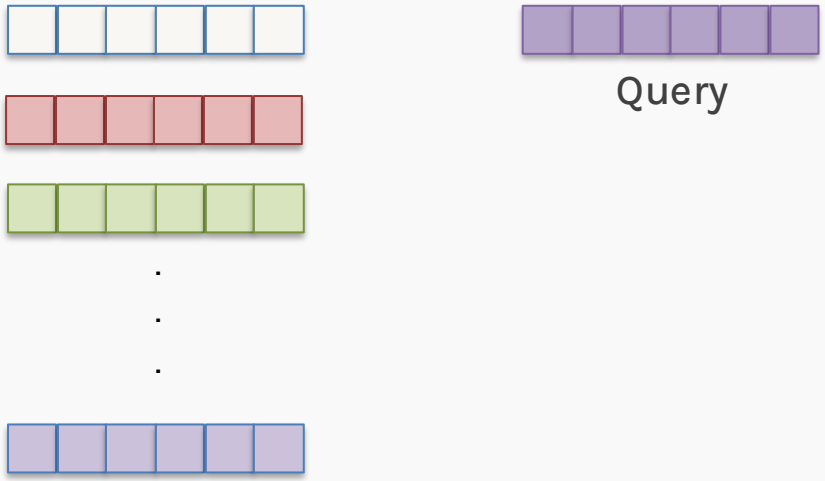
Vectors capture the essential features of the original data in a high-dimensional space.

Naïve RAG – Vector Database

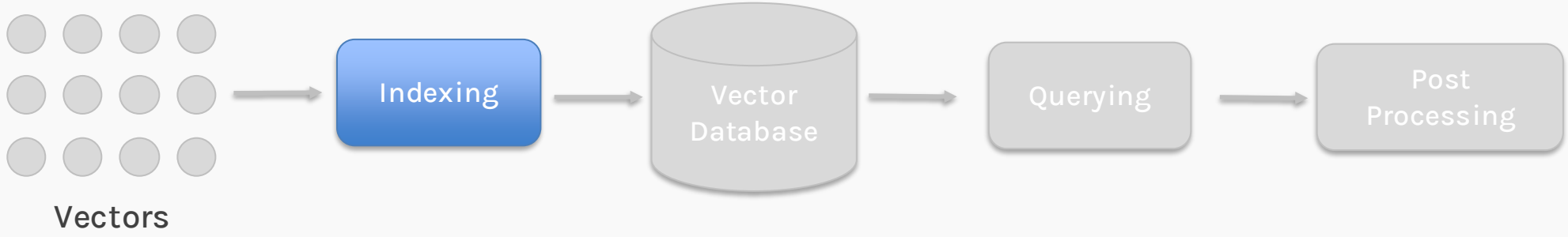




In reality, we could have millions of vectors to deal with.

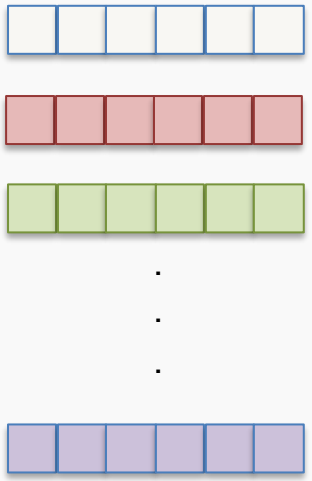


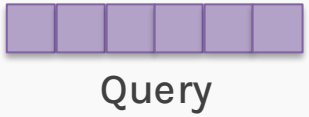
Naïve RAG – Vector Database



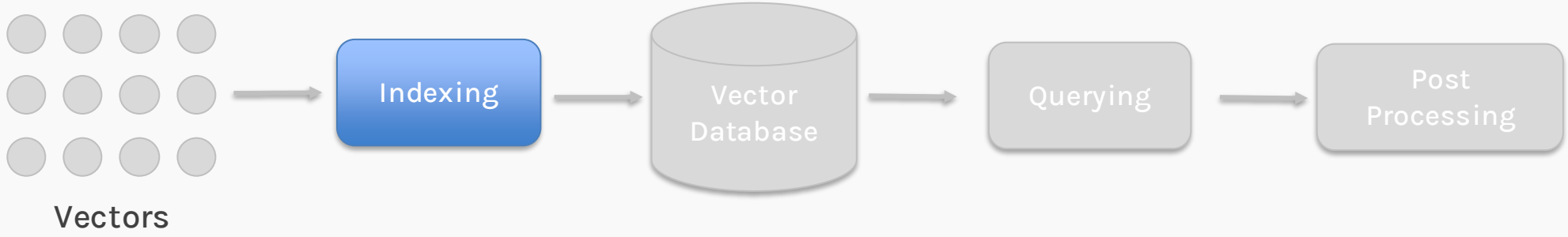


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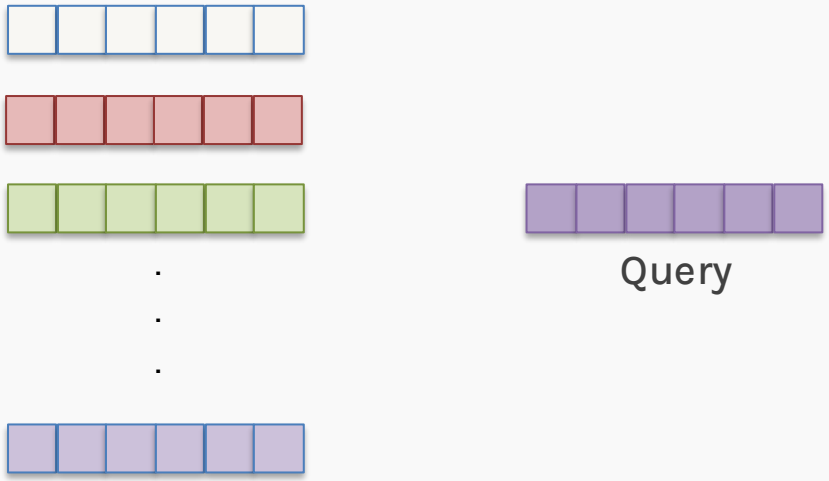


Naïve RAG – Vector Database

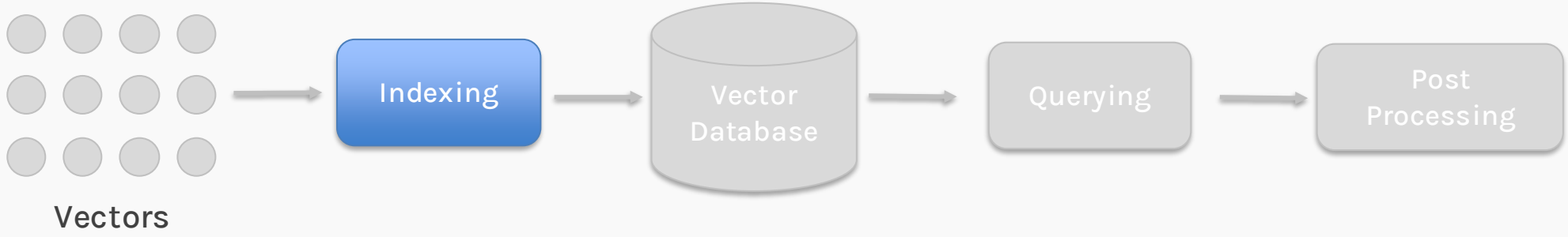




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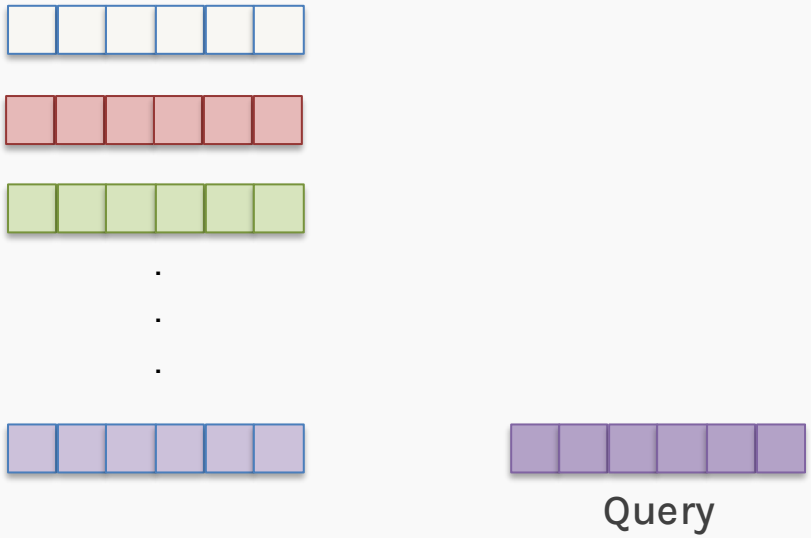


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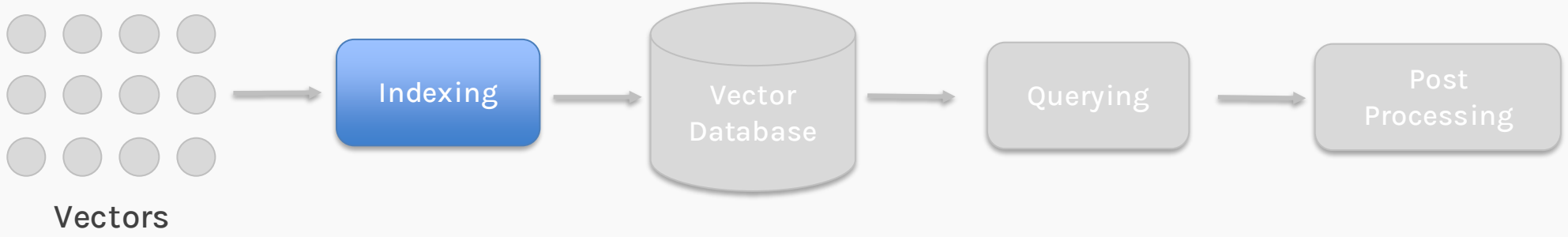


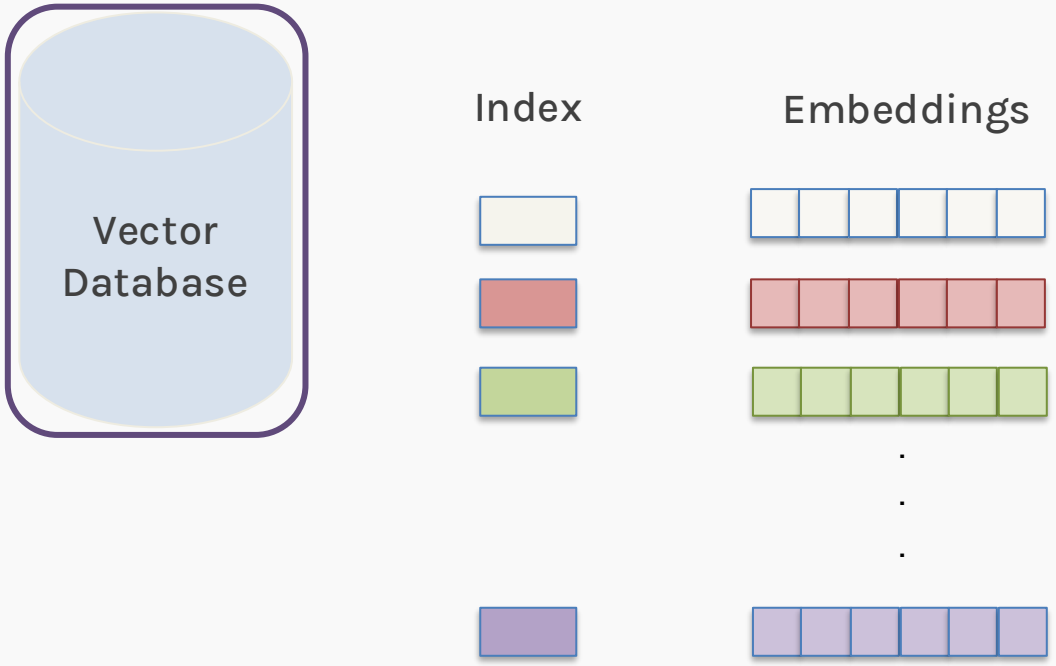
In reality, we could have millions of vectors to deal with.



Comparing them sequentially would be a very slow process.

Naïve RAG – Vector Database

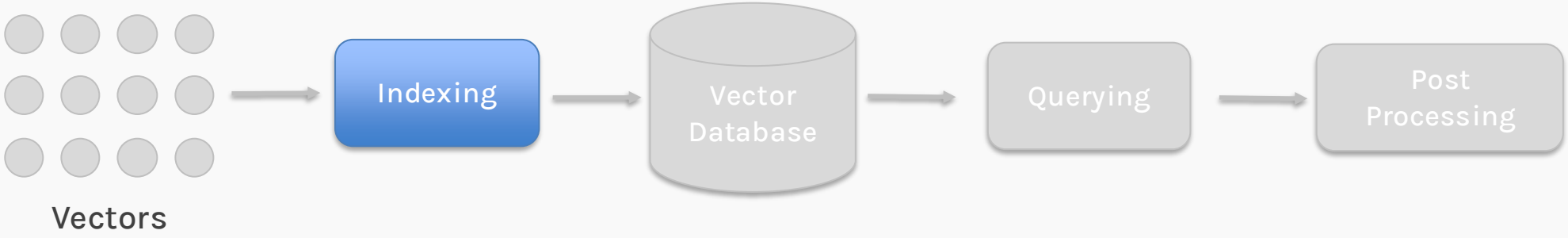




Indexing is the process of mapping the vectors to a data structure that will enable us to search faster.

When we query for a particular vector, the index helps us quickly zoom into a small subset of vectors that are likely to be similar.

Naïve RAG – Vector Database



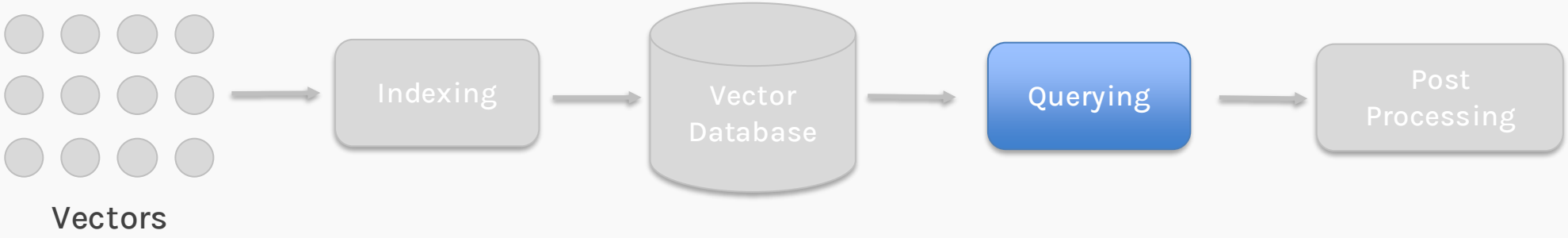


We usually use one of the three algorithms to do indexing:

1. Hashing (Locality Sensitive Hashing - LSH)
2. Quantization (Product Quantization - PQ)
3. Graph Based (Hierarchical Navigable Small World - HNSW)

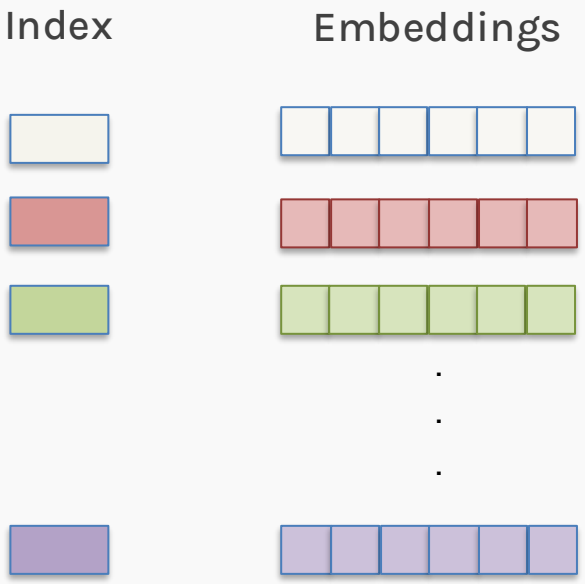
Nowadays, most relational databases like PostgreSQL have introduced vector as a column datatype, providing vector database benefits within traditional databases.

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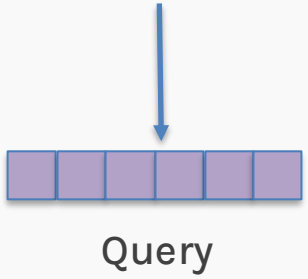




When querying, the vector database compares the indexed vectors to the query vector to determine the nearest vector neighbor.



Query: What are different techniques of cheese making?



Pavlos game #4275



WHO IS MOST SIMILAR TO PAVLOS?

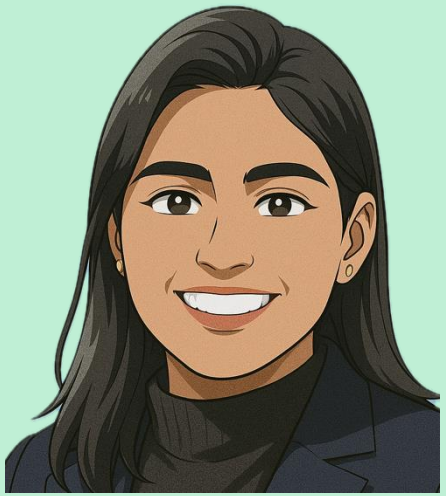
Option a



Option B



Option C



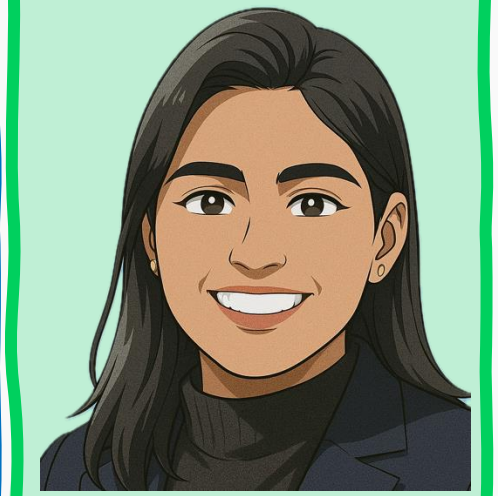
BIG 5

Personality test
RESULTS





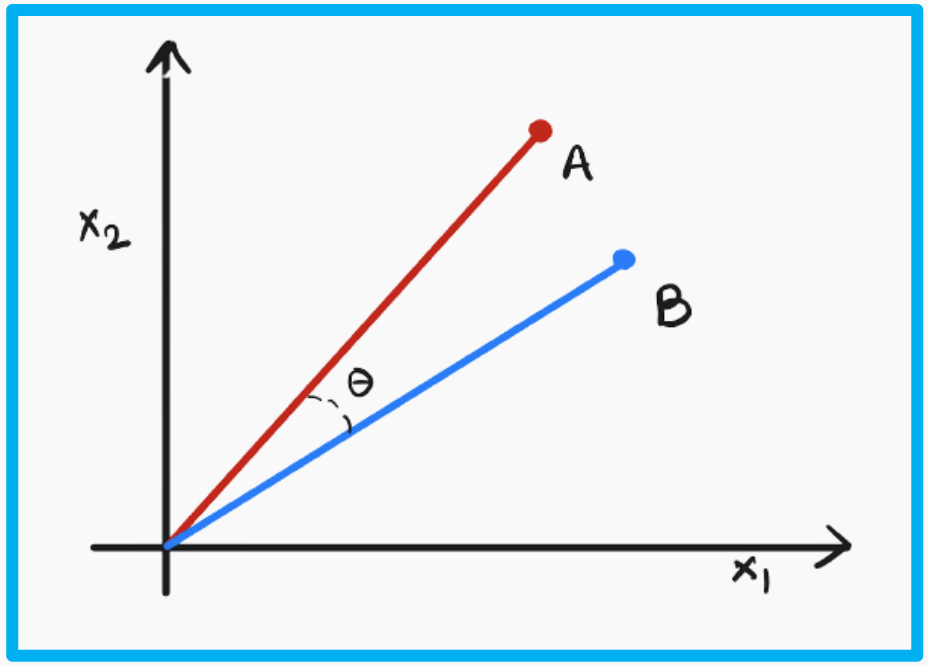




EXTROVERSION	4.8	3.3	4.8	1.0
Emotional stability	4.5	4.0	2.7	3.6
agreeableness	4.2	4.1	4.9	3.8
conscientiousness	3.3	3.1	2.5	4.4
imagination	4.1	4.6	4.2	2.2

USING PERSONALITY DATA TO FIND SIMILARITY

What is “Cosine Similarity”?

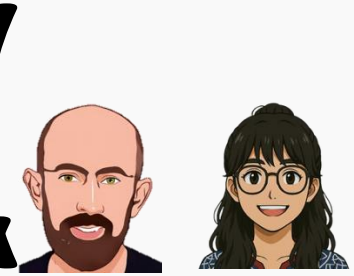


$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

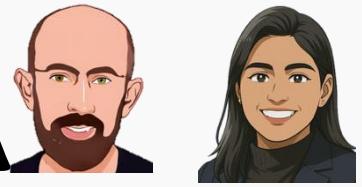
Where A_i & B_i are components of vector A & B respectively



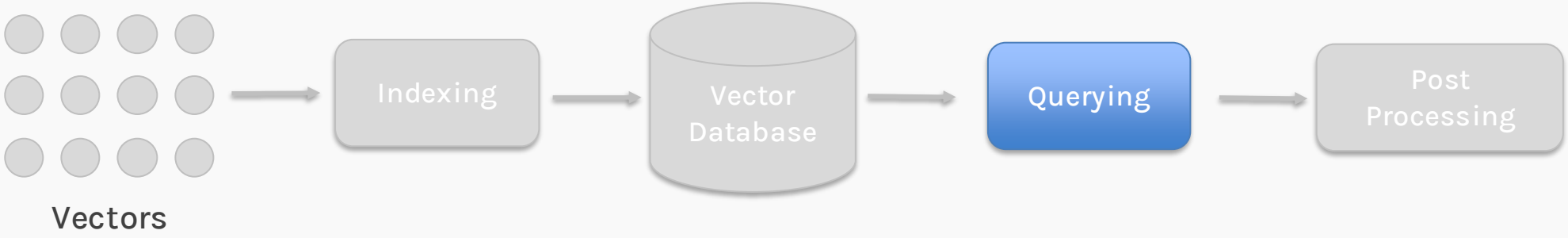
WHO IS MOST SIMILAR TO PAVLOS?

COSINE SIMILARITY  = 0.986 ✓

COSINE SIMILARITY  = 0.976

COSINE SIMILARITY  = 0.886

Naïve RAG – Vector Database





When querying, the vector database compares the indexed vectors to the query vector to determine the nearest vector neighbor.

But, how do we compare?

We can use one of the following similarity measures to find the nearest neighbor:

1. Cosine Similarity
2. Euclidian Distance
3. Dot Product

How do these measures differ from each other?

Similarity Measures

Measure	Formula	What is Measures	Range	Strengths	When to Use
Cosine Similarity	$\frac{x.y}{\|x\|\|y\|}$	Cosine of the angle between two vectors (orientation).	[-1, 1] (-1: opposite, 1: identical).	Focuses on direction, ignores magnitude.	- Use for **text embeddings** (e.g., semantic search). - Normalized embeddings, LLM-based retrieval (e.g., BERT/GPT).

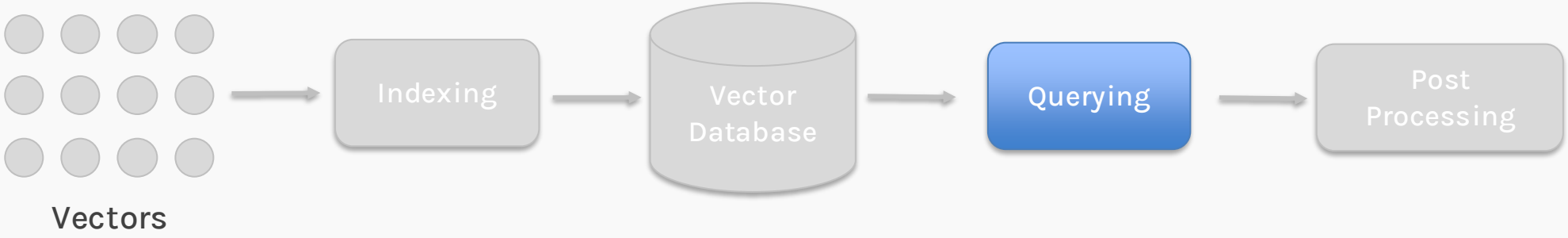
Similarity Measures

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Euclidean Distance	$\sqrt{(x-y)^2}$	Straight-line distance between two points.	[0, ∞] (0: identical vectors).	Captures magnitude differences well.	- Use when working with **numeric feature vectors** (e.g., metadata or tabular data). - Structured or multi-dimensional data where **magnitude** matters.
		Product of magnitudes and	Unbounded (positive:	Combines magnitude and	- Use for **ranking documents** based on relevance scores

Similarity Measures

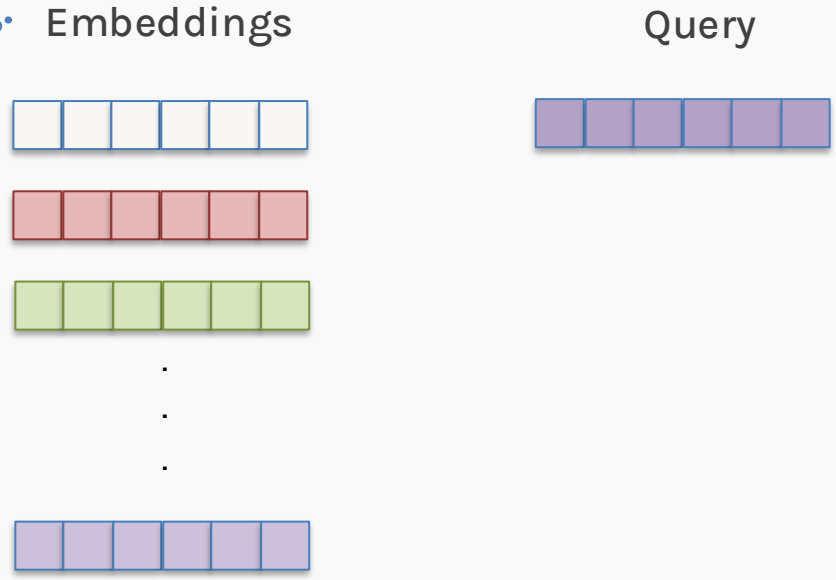
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Euclidean Distance	$\sqrt{(x-y)^2}$	Straight-line distance between two points.	[0, ∞] (0: identical vectors).	Captures magnitude differences well.	- Use when working with **numeric feature vectors** (e.g., metadata or tabular data). - Structured or multi-dimensional data where **magnitude** matters.
Dot Product	$x.y$	Product of magnitudes and cosine of the angle.	Unbounded (positive: similar, negative: dissimilar).	Combines magnitude and direction effectively.	- Use for **ranking documents** based on relevance scores. - Ideal when the magnitude of the vector represents **importance or confidence** (e.g., user preference weights).

Naïve RAG – Vector Database

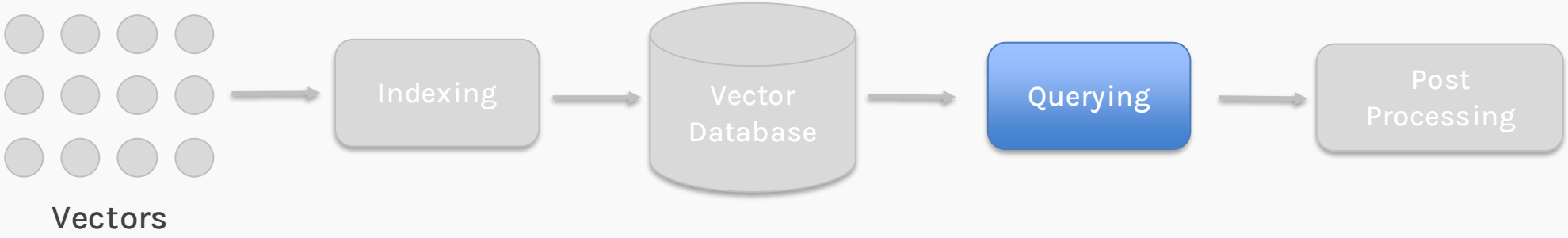




This leads to a ranking of embeddings based on their similarity with the query embedding.

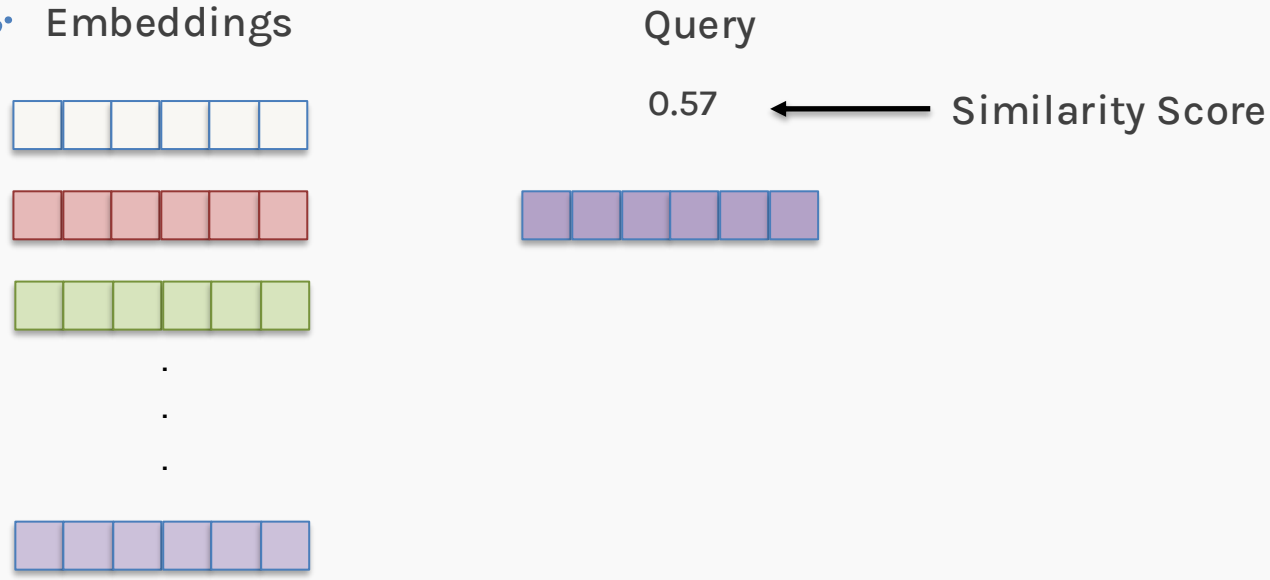


Naïve RAG – Vector Database

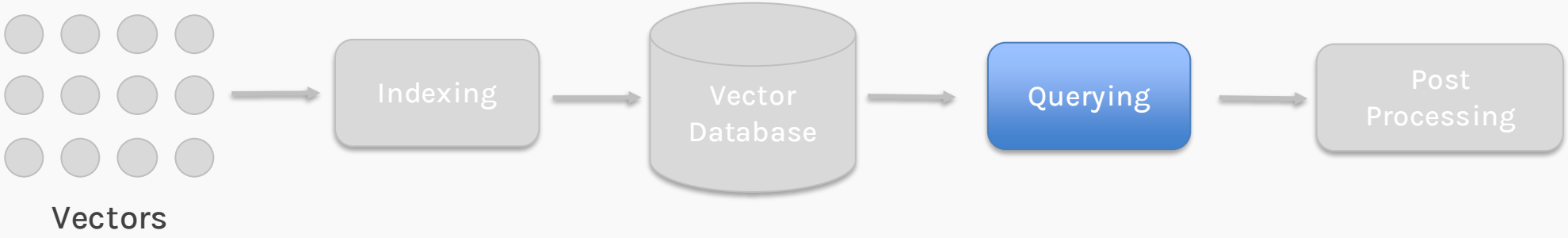




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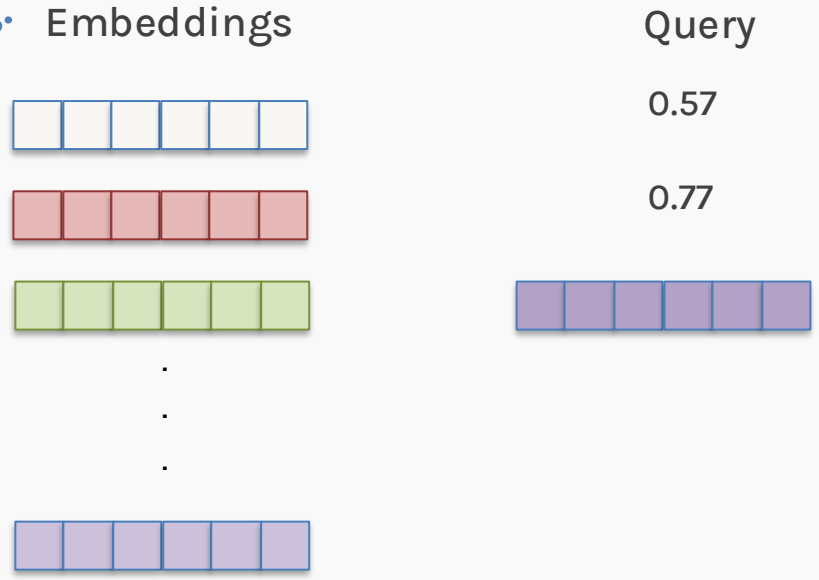


Naïve RAG – Vector Database

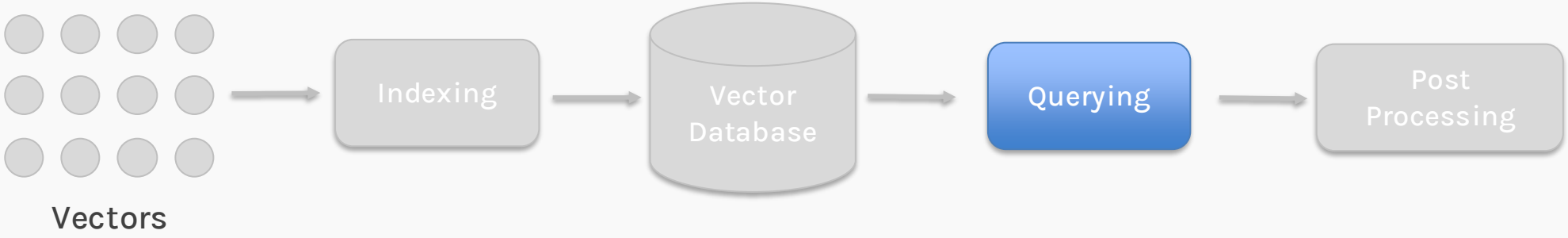




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Naïve RAG – Vector Database

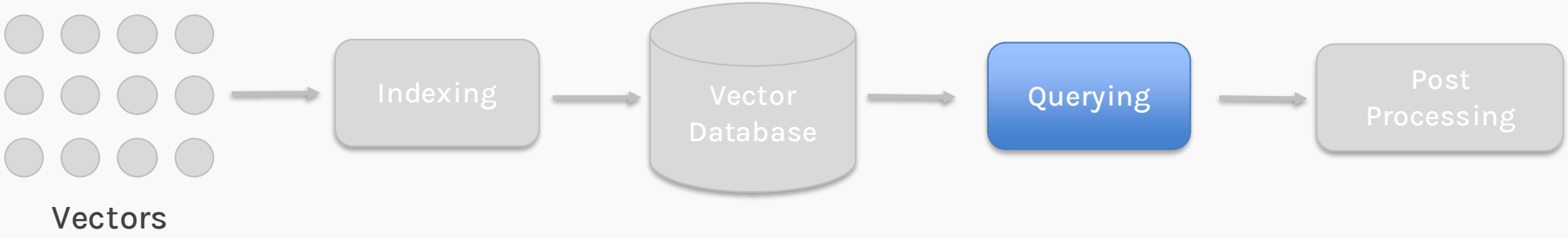




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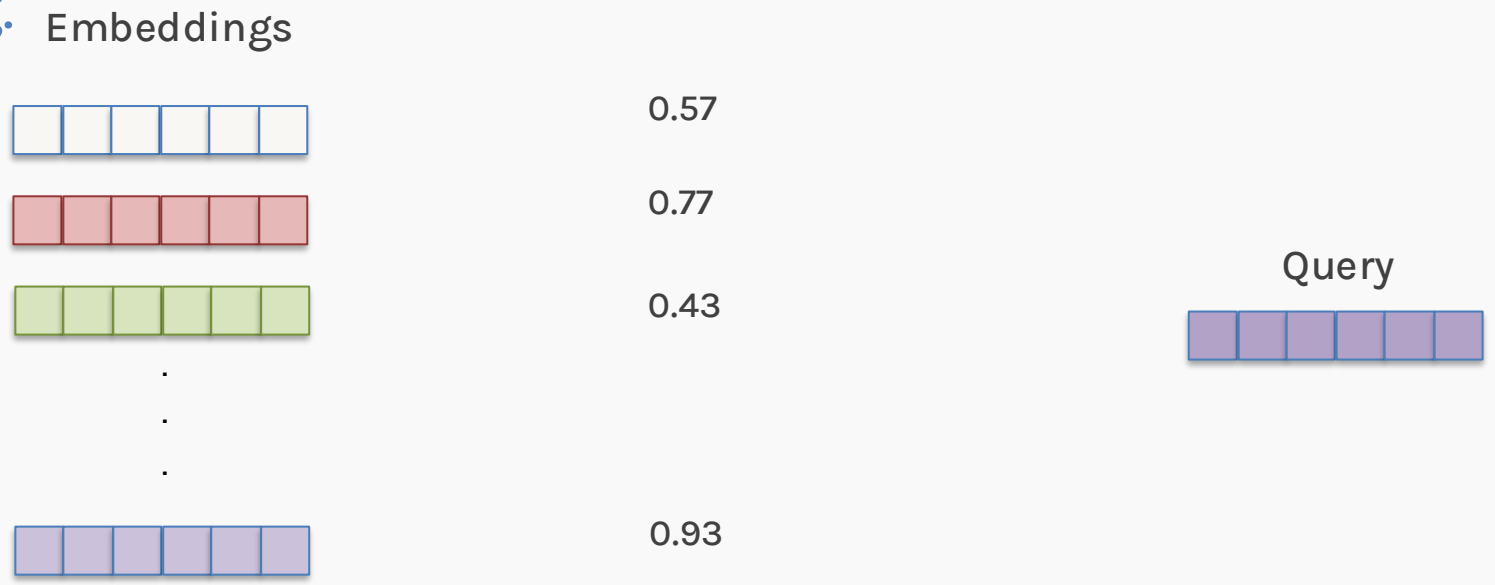


Naïve RAG – Vector Database

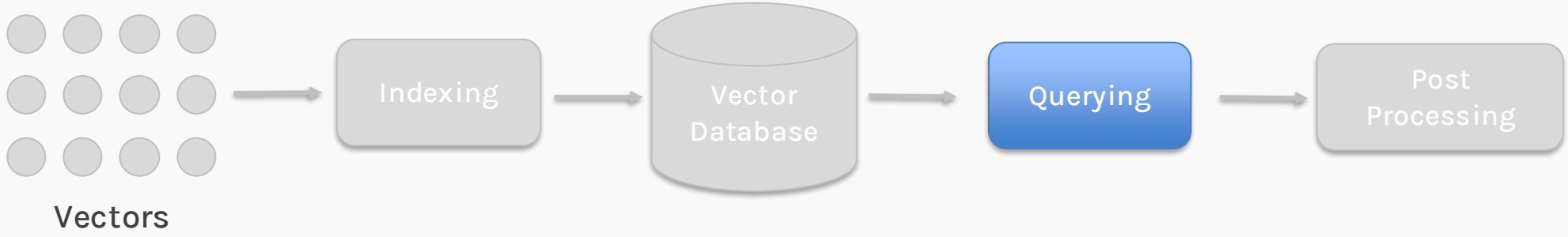




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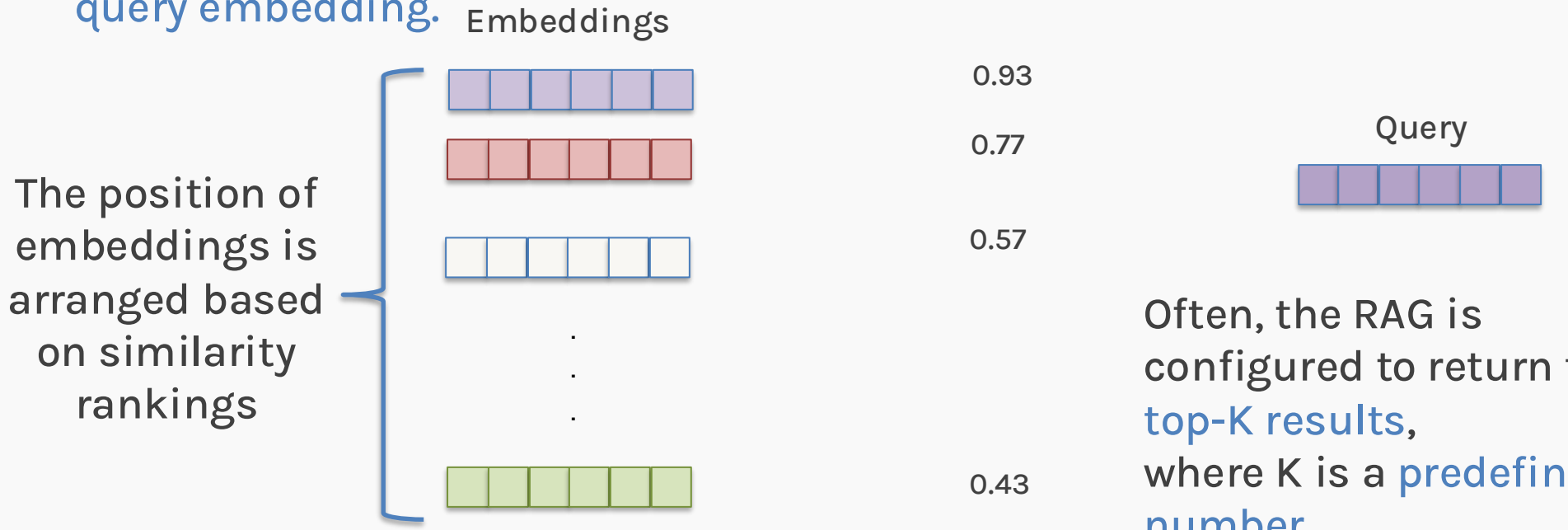


Naïve RAG – Vector Database



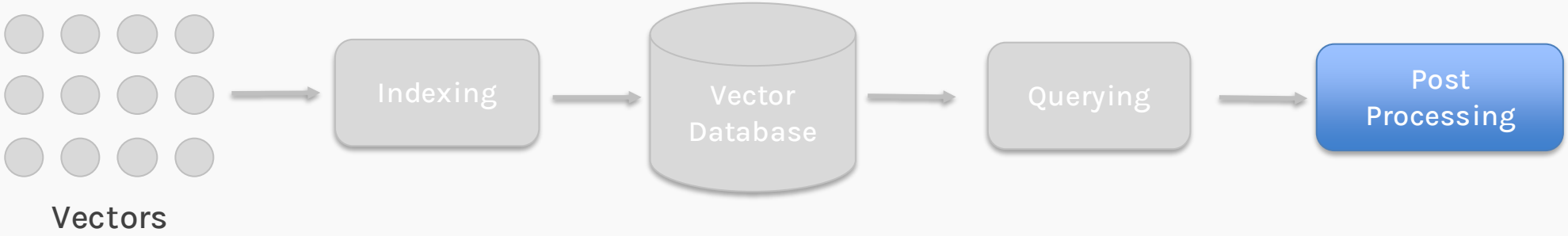


This leads to a ranking of embeddings based on their similarity with the query embedding.



Often, the RAG is configured to return the top-K results, where K is a predefined number.

Naïve RAG – Vector Database





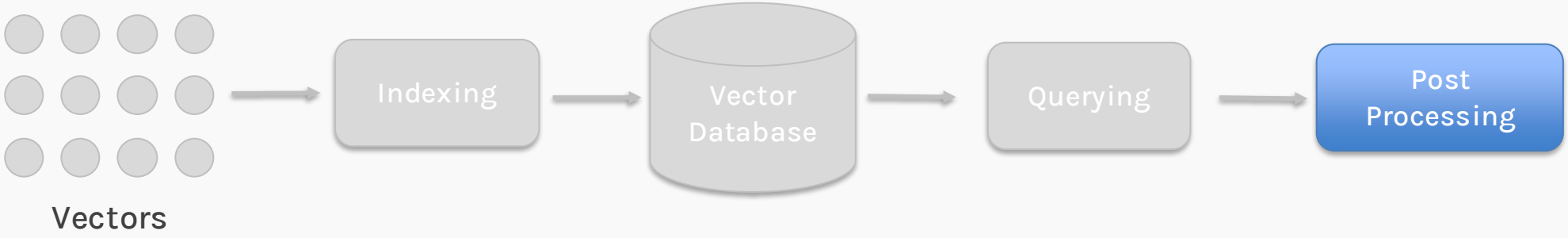
In advanced RAG, an additional pre-processing step re-ranks the retrieved results using a different similarity measure, improving the quality of inputs for the next stage.

We could also do some metadata filtering here.

Note: *This is an* ***optional*** *step.*

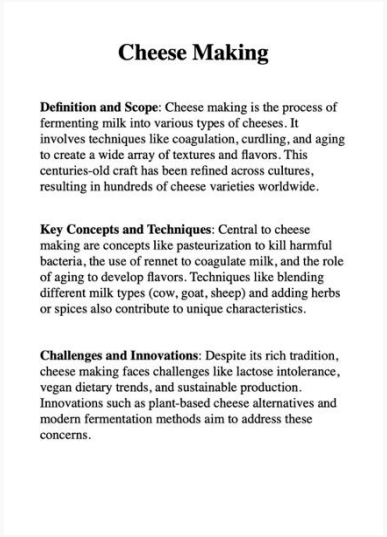
Metadata is data that provides information about other data (in our case, the vectors).

Naïve RAG – Vector Database





Metadata filtering helps refine and narrow down the search results based on metadata.



Cheese Making

Definition and Scope: Cheese making is the process of fermenting milk into various types of cheeses. It involves techniques like coagulation, curdling, and aging to create a wide array of textures and flavors. This centuries-old craft has been refined across cultures, resulting in hundreds of cheese varieties worldwide.

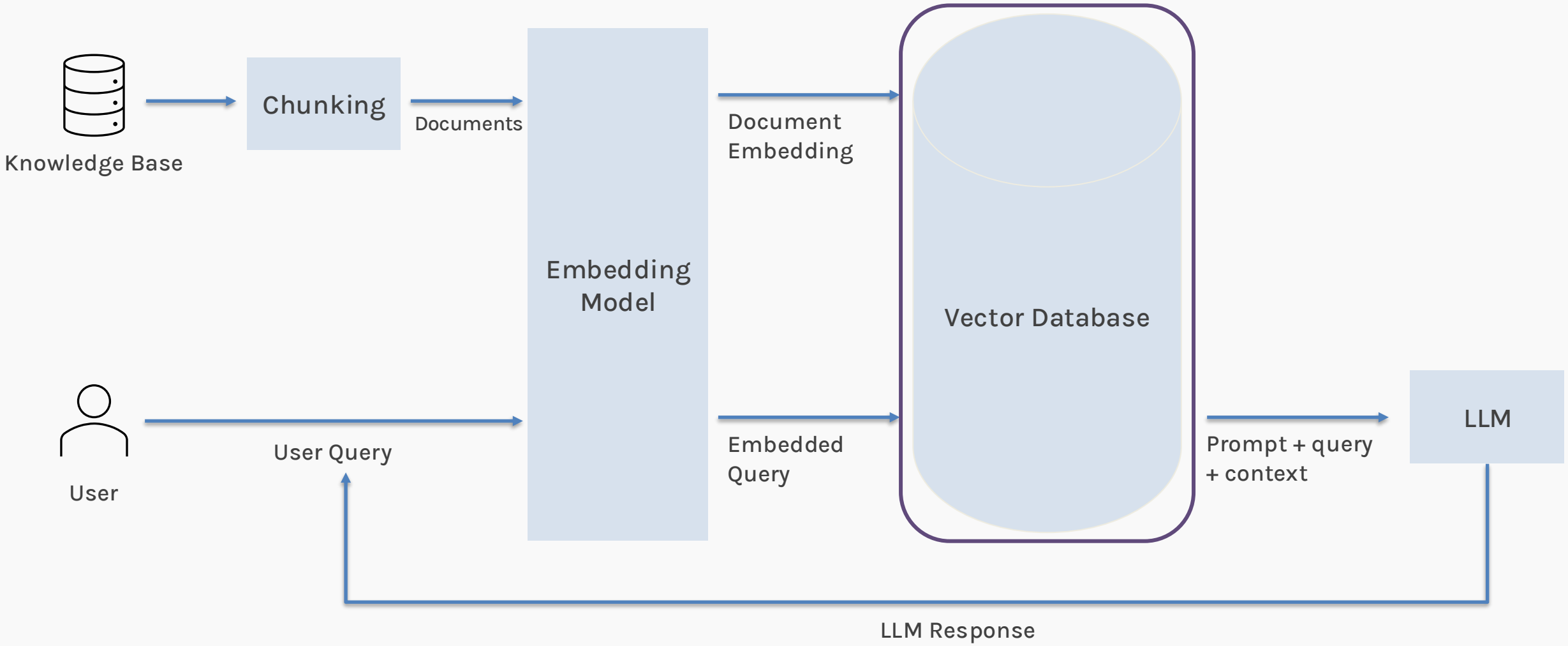
Key Concepts and Techniques: Central to cheese making are concepts like pasteurization to kill harmful bacteria, the use of rennet to coagulate milk, and the role of aging to develop flavors. Techniques like blending different milk types (cow, goat, sheep) and adding herbs or spices also contribute to unique characteristics.

Challenges and Innovations: Despite its rich tradition, cheese making faces challenges like lactose intolerance, vegan dietary trends, and sustainable production. Innovations such as plant-based cheese alternatives and modern fermentation methods aim to address these concerns.

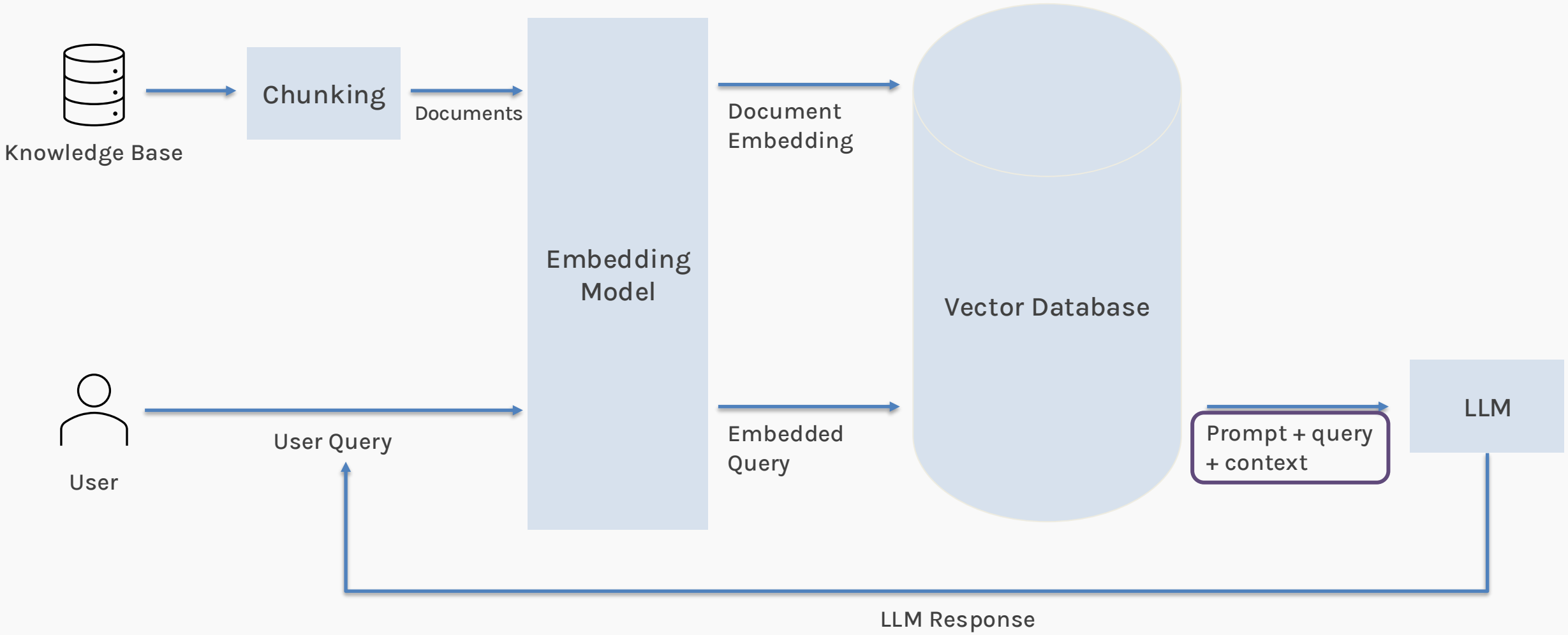
For this data, the metadata may be:

1. Author
2. Publication date
3. Category

Naïve RAG


Knowledge Base
Chunking
Documents
Embedding Model
Document Embedding
Vector Database
User
User Query
Embedded Query
Prompt + query + context
LLM
LLM Response

Naïve RAG


Knowledge Base
Chunking
Documents
Embedding Model
Document Embedding
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User Query
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Prompt + query + context
LLM
LLM Response

Naïve RAG – Prompt + query + context

Prompt

System prompts acts as an instruction given to the model to guide its behavior and responses

Query

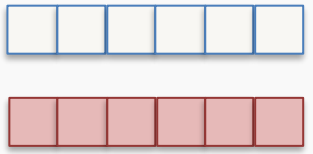
The query is the user's initial input.



Query

Context

Context refers to the information retrieved from the vector database that is relevant to the query,



Context

Naïve RAG – Prompt + query + context

Let's see an example of how this would look like:

System Prompt

"You are a Cheese Expert. When answering user questions explain in detail the key techniques and concepts involved in cheese making"

User Query

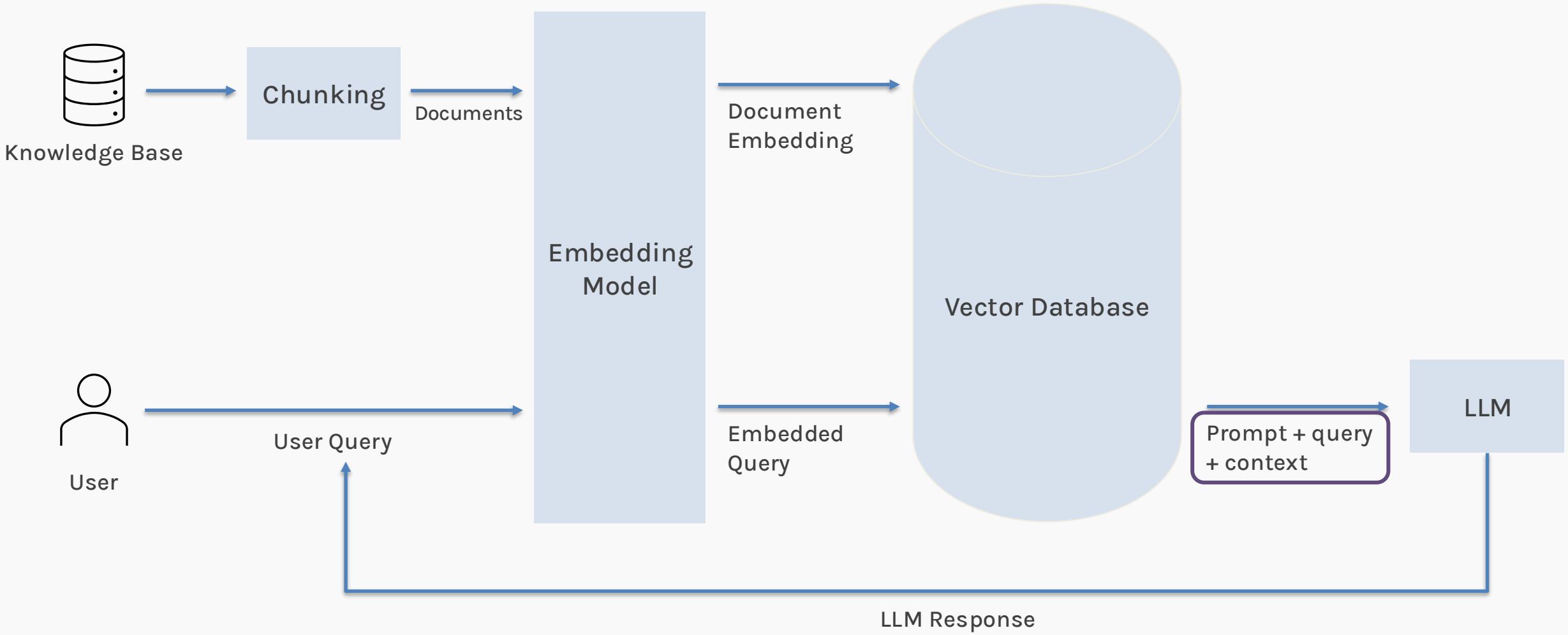
"What are the main steps and methods used in the cheese making process?"

Context (Relevant Extracts)

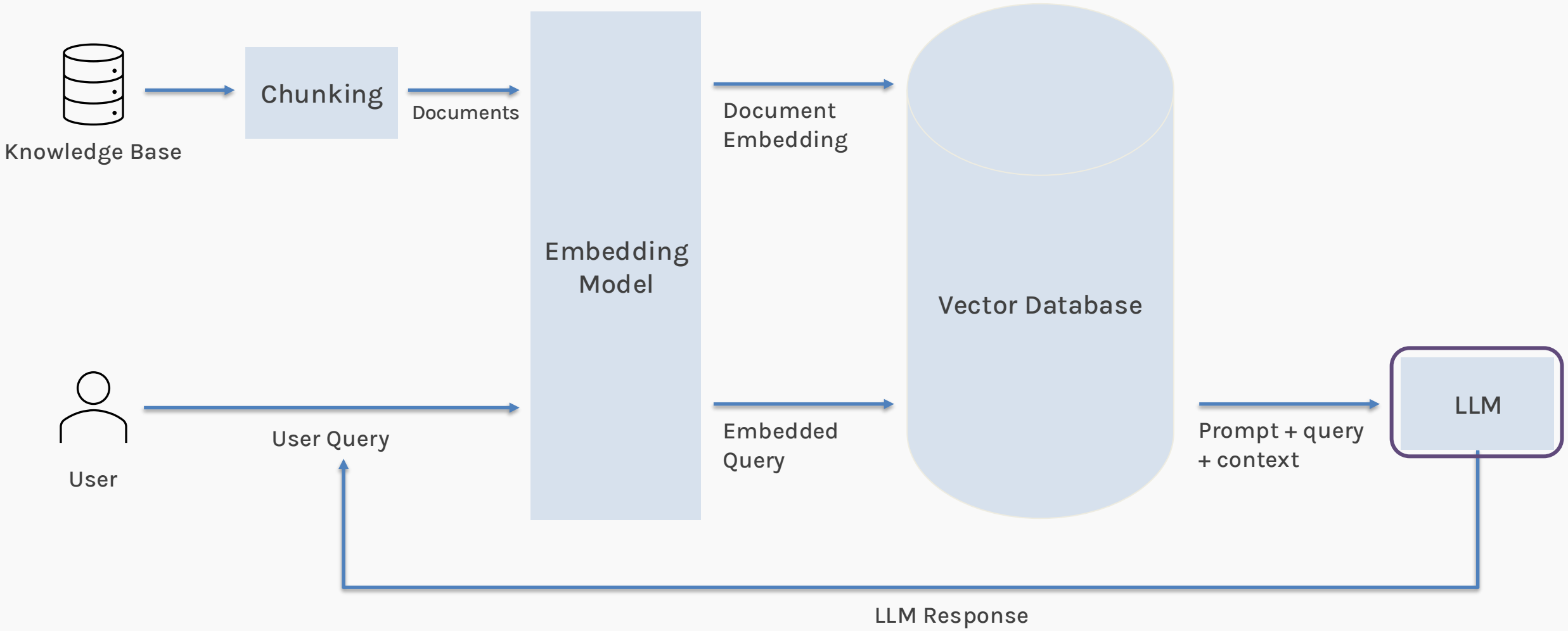
Central to cheese making are concepts like pasteurization, the use of rennet to coagulate milk, and the role of aging in developing flavors. Techniques such as blending different milk types (cow, goat, sheep) and adding herbs/spices contribute to unique cheese characteristics.

Final Prompt

Naïve RAG


Knowledge Base
Chunking
Documents
Embedding Model
Document Embedding
Vector Database
User
User Query
Embedded Query
Prompt + query + context
LLM
LLM Response

Naïve RAG

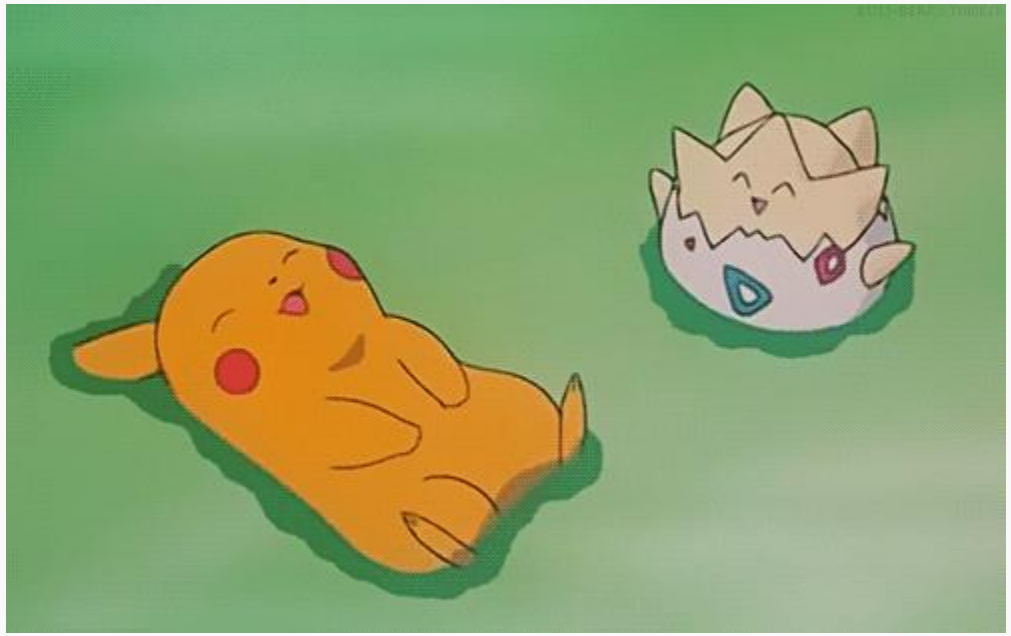

Knowledge Base
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Naïve RAG – Vector Database

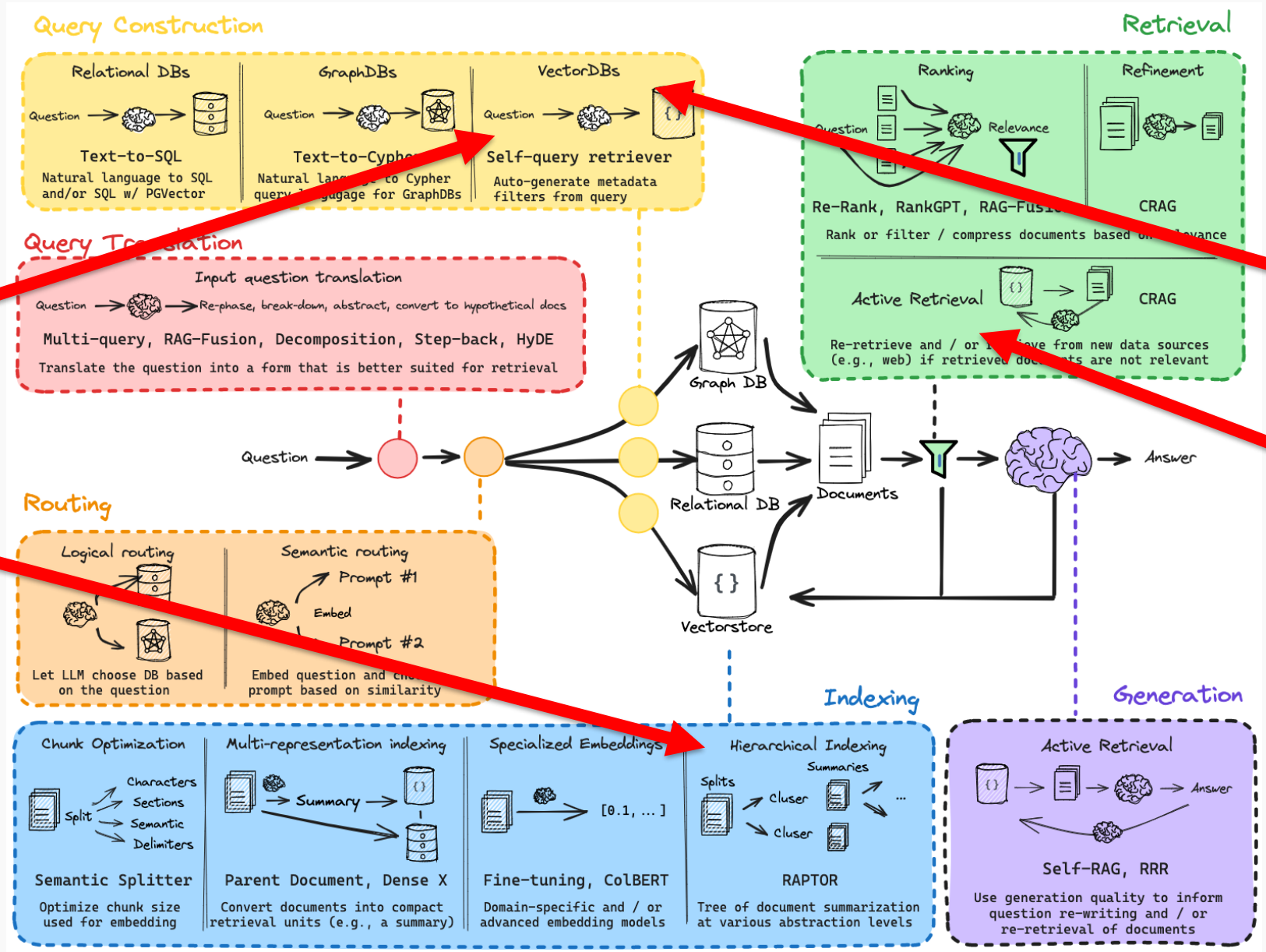
LLM

Well, we have already talked about LLMs in the previous sessions.

Nothing new to be added.



Retrieval Augmented Generation



~~What is Vector Database?~~

~~What is query~~

~~What is embedding?~~

~~What is context?~~



THANK YOU!